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**Methods for determination of sulfur
oxides in flue gas**

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Foreword

This translation has been made based on the original Japanese Industrial Standard revised by the Minister of Economy, Trade and Industry through deliberations at the Japanese Industrial Standards Committee as the result of proposal for revision of Japanese Industrial Standard submitted by Japan Environmental Measurement & Chemical Analysis Association (JEMCA)/Japanese Standards Association (JSA) with the draft being attached, based on the provision of Article 12 Clause 1 of the Industrial Standardization Law applicable to the case of revision by the provision of Article 14.

Consequently, JIS K 0103: 2005 is replaced with this Standard.

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Methods for determination of sulfur oxides in flue gas

Introduction

This Japanese Industrial Standard has been prepared based on the first edition of ISO 7934 published in 1989 and the first edition of ISO 11632 published in 1998. For ISO 7934, the technical contents thereof have been specified in Annex A without any modifications. For ISO 11632, the specification items and specification contents not given in it have been added but part of it has not been adopted.

The portions given sidelines or dotted underlines are the matters in which the contents of the corresponding International Standards have been modified. A list of modifications with the explanations is given in Annex JE. Annex JA to Annex JD are the matters not contained in the corresponding International Standards.

1 Scope

This Standard specifies the methods for analysis of sulfur oxides in flue gas (hereafter referred to as "analysis methods").

In this Standard, the flue gas means the gas which is exhausted into flue, chimney, duct, etc. from such processes as burning of fuels, roasting of ores, refining of metals, manufacturing of sulfur, other chemical reactions, or desulfurization.

NOTE : The International Standards corresponding to this Standard and the symbol of degree of correspondence are as follows.

ISO 7934 : 1989 *Stationary source emissions — Determination of the mass concentration of sulfur dioxide — Hydrogen peroxide/barium perchlorate/Thorin method*

ISO 11632 : 1998 *Stationary source emissions — Determination of mass concentration of sulfur dioxide — Ion chromatography method*
(Overall evaluation : MOD)

In addition, symbols which denote the degree of correspondence in the contents between the relevant International Standards and JIS are IDT (identical), MOD (modified), and NEQ (not equivalent) according to ISO/IEC Guide 21-1.

2 Normative references

The following standards contain provisions which, through reference in this text, constitute provisions of this Standard. The most recent editions of the standards (including amendments) indicated below shall be applied.

JIS B 7981 *Automated measuring systems and analyzers for sulfur dioxide in flue gas*

NOTE : Corresponding International Standard : ISO 7935 : 1992 *Stationary source emissions — Determination of the mass concentration of sulfur dioxide — Performance characteristics of automated measuring methods* (MOD)

JIS K 0050 *General rules for chemical analysis*

JIS K 0055 *General rules for calibration method of gas analyzer*

JIS K 0095 *Methods for sampling of flue gas*

NOTE : Corresponding International Standard : ISO 10396 : 1993 *Stationary source emissions — Sampling for the automated determination of gas concentrations (MOD)*

JIS K 0104 *Methods for determination of nitrogen oxides in flue gas*

JIS K 0107 *Methods for determination of hydrogen chloride in flue gas*

JIS K 0115 *General rules for molecular absorptiometric analysis*

JIS K 0127 *General rules for ion chromatographic analysis*

JIS K 0557 *Water used for industrial water and wastewater analysis*

JIS K 8005 *Reference materials for volumetric analysis*

JIS K 8032 *Acetonitrile (Reagent)*

JIS K 8102 *Ethanol (95) (Reagent)*

JIS K 8150 *Sodium chloride (Reagent)*

JIS K 8155 *Barium chloride dihydrate (Reagent)*

JIS K 8180 *Hydrochloric acid (Reagent)*

JIS K 8230 *Hydrogen peroxide (Reagent)*

JIS K 8295 *Glycerol (Reagent)*

JIS K 8355 *Acetic acid (Reagent)*

JIS K 8374 *Lead (II) acetate trihydrate (Reagent)*

JIS K 8376 *Barium acetate (Reagent)*

JIS K 8576 *Sodium hydroxide (Reagent)*

JIS K 8603 *Soda lime (Reagent)*

JIS K 8622 *Sodium hydrogen carbonate (Reagent)*

JIS K 8625 *Sodium carbonate (Reagent)*

JIS K 8839 *2-Propanol (Reagent)*

JIS K 8844 *Bromophenol blue (Reagent)*

JIS K 8863 *Boric acid (Reagent)*

JIS K 8866 *Sodium tetraborate decahydrate (Reagent)*

JIS K 8896 *Methyl red (Reagent)*

JIS K 8897 *Methylene blue (Reagent)*

JIS K 8951 *Sulfuric acid (Reagent)*

JIS K 8962 *Potassium sulfate (Reagent)*

JIS K 8987 *Sodium sulfate (Reagent)*

JIS K 9524 *Arsenazo III (Reagent)*

JIS K 9704 *2-Amino-2-hydroxymethyl-1,3-propanediol (Reagent)*

JIS K 9808 *2-Bis (2-hydroxyethyl) amino-2-hydroxymethyl-1,3-propanediol (Bis-Tris)*

JIS R 3503 *Glass apparatus for chemical analysis*

JIS Z 8808 *Methods of measuring dust concentration in flue gas*

ISO 3696 *Water for analytical laboratory use — Specification and test methods*

3 General matters

The general matters shall be as follows.

- a) General matters common to the chemical analysis shall be in accordance with JIS K 0050.
- b) General matters common to the sampling of flue gas shall be in accordance with JIS K 0095.
- c) General matters common to the ion chromatography shall be in accordance with JIS K 0127.
- d) General matters common to the molecular absorptiometric analysis shall be in accordance with JIS K 0115.
- e) Water to be used in analysis shall be of A2 or A3 specified in clause 4 of JIS K 0557 or any water equivalent to these.
- f) Reagents to be used shall be of the highest grade or of an appropriate quality as specified in a relevant Japanese Industrial Standard, if applicable. If the reagent does not have a relevant specification in Japanese Industrial Standard, that of a quality adequate for the analysis shall be used.
- g) For the reference solution, the reference solution with traceability secured or that solution diluted to a definite concentration, and in addition, the solutions of which the preparation methods are specified for respective test items shall be used.
- h) Apparatus and instrument to be used shall be provided with the specified functions.
- i) When disposing flue gas, or absorbing solution of the flue gas used for the analysis of hydrogen sulfide, it shall be done with utmost care.

NOTE : For the reagents with traceability secured, there are the reagents labelled with JCSS mark specified in the national measurement standard (Article 134 of the Measurement Act).

4 Classification and summary of analysis methods

The classification and summary of analysis methods shall be as follows.

- a) **Chemical analysis method (objective components : SO₂ + SO₃)** The classification and summary of chemical analysis method shall be given in table 1.

Table 1 Type and summary of chemical analysis method

Type of analysis method	Summary			Applicable condition
	Outline	Sampling	Determination range vol ppm (mg/m ³)	
Ion chromatography	After absorption of sulfur oxides in sample gas into hydrogen peroxide solution to make sulfuric acid, it is introduced into ion chromatograph and chromatogram thereof is recorded.	Absorbing bottle method Absorption solution : hydrogen peroxide solution (1 + 99) or (1 + 9) Liquid amount : 25 ml ^{a)} × 2 bottles or 50 ml ^{b)} × 2 bottles Standard sampling volume : 20 L	0.5 to 58 ^{c)} (1.4 to 160) ^{c)} 12 to 290 ^{d)} (34 to 830) ^{d)}	According to 7.1.1
Precipitation titrimetry (arsenazo III method)	After absorption of sulfur oxides in sample gas into hydrogen peroxide solution to make sulfuric acid, add 2-propanol and acetic acid, and titrate it with barium acetate solution with adding arsenazo III as an indicator.	Absorbing bottle method Absorption solution : hydrogen peroxide solution (1 + 9) Liquid amount : 50 ml ^{b)} × 2 bottles Standard sampling volume : 20 L	140 to 700 ^{d)} (400 to 2 000) ^{d)} For photometric titrimetry, lowest determination limit : 50 (about 140)	Not specified
Notes	^{a)} Liquid amount for absorption when using absorbing bottle (100 ml in capacity). ^{b)} Liquid amount for absorption when using absorbing bottle (250 ml in capacity). ^{c)} In the case of employing the sample solution for analysis prepared by diluting 50 ml of absorption solution through which the sample gas has been passed to 100 ml. The determination range shown in this column is obtained according to the standard sampling volume of the sample gas, the amount of sample solution for analysis and the optimum range of the working curve. When determining the concentration in this determination range or under, the concentrating column shall be used. ^{d)} In the case of employing the sample solution for analysis prepared by diluting 100 ml of absorption solution through which the sample gas has been passed to 250 ml. The determination range shown in this column is obtained according to the standard sampling volume of the sample gas, the amount of sample solution for analysis and the optimum range of the working curve. For the concentration over the determination range, it shall be measured in such a manner that after collecting by using aqueous solution of hydrogen peroxide solution (1 + 9), the sample solution for analysis shall be diluted within the determination range.			

NOTE 1 The symbols, vol ppm and mg/m³, in this Standard express the volume fraction and the mass concentration, respectively under the standard condition [273.15 K (0 °C), 101.32 kPa].

NOTE 2 For the simplified method for analysis for the concentration of sulfur dioxide in flue gas, a detecting tube method, a simplified type of automated measuring instrument, and so on are included. For the detecting tube, the sample gas of approximately 100 ml is sucked in the detecting tube, reacted with the reagent in the tube, and the concentration of sulfur dioxide is obtained from the coloured length of reagent. The measurement range of concentration is 5 vol ppm to 3 % by the suction of 100 ml, depending on the type of detecting tube and the suction amount of sample gas. The concentration of sulfur dioxide in flue gas can be approximated by means of this method.

Besides the methods in table 1, the precipitation titrimetry (Thorin method) of ISO 7934 is specified in Annex A (normative), the turbidimetry (light scattering method) is specified in Annex JA (normative), the neutralization titrimetry is specified in Annex JB (normative), the simultaneous analysis method of sulfur oxides and hydrogen chloride by ion chromatography is specified in Annex JC (normative) and the simultaneous analysis method of sulfur oxides, hydrogen chloride and nitrogen oxides by ion chromatography is specified in Annex JD (normative).

- b) **Automated measuring method (objective component : SO₂)** The automated measuring method for sulfur dioxide in flue gas shall be in accordance with the method specified in JIS B 7981.

5 Sampling method of sample gas

5.1 General

The sampling method of sample gas used in the chemical analysis shall be as follows. The sampling position of sample gas used for analysis shall be selected from the point where representative sample gas can be sampled. The sample gas shall be taken at least twice without interval as far as possible, as a rule, at the same sampling position. Each of them shall be used for the analysis.

5.2 Reagents and preparation of absorption solution

5.2.1 Reagents

Hydrogen peroxide That specified in JIS K 8230.

5.2.2 Preparation of absorption solution

- a) **Hydrogen peroxide solution (1 + 9)** Take 100 ml of hydrogen peroxide and add 900 ml of water to it. Store it in dark and cold place.
- b) **Hydrogen peroxide solution (1 + 99)** Take 10 ml of hydrogen peroxide and add 990 ml of water to it. Store it in dark and cold place.

5.3 Implements and apparatus

a) Absorbing bottle

- 1) **Case of ion chromatography** Connect two absorbing bottles (100 ml or 250 ml capacity) as exemplified in figure 1.
- 2) **Case of precipitation titrimetry (arsenazo III method)** Connect two absorbing

bottles (250 ml capacity) as exemplified in figure 1.

b) **Sampling apparatus of sample gas** The construction is as exemplified in figure 2, and the following conditions shall be satisfied.

- 1) The sampling tube of sample gas (B) shall not be eroded by the corrosive gas in flue gas, and shall be made of the material such as the glass tube, quartz tube, tetrafluoro ethylene resin tube which does not absorb the corrosive gas.
- 2) For preventing the dust contamination into the sample gas, the end or other suitable place of sampling tube of sample gas (B) shall be filled with appropriate filtering medium ¹⁾.
- 3) For preventing the moisture condensation in the sample gas, the apparatus shall be so constructed that the part from the sampling tube to the three-way cock for flow path switch (P₁) can be heated.

To connect these parts, the ground connecting pipes, silicone rubber pipes or polytetrafluoroethylene pipes shall be used. The piping shall be as short as possible, and if there is a possibility of moisture condensation, the part between the sampling tube and the cock (P₁) shall be heated at about 160 °C.

- 4) The whole apparatus shall be set up so as to get rid of any leakage at each connection.

Note ¹⁾ The material, such as silica wool, non-alkaline wool which causes no chemical reaction with the component in flue gas shall be used.

Unit : mm Inside () for 100 ml and
outside () for 250 ml.

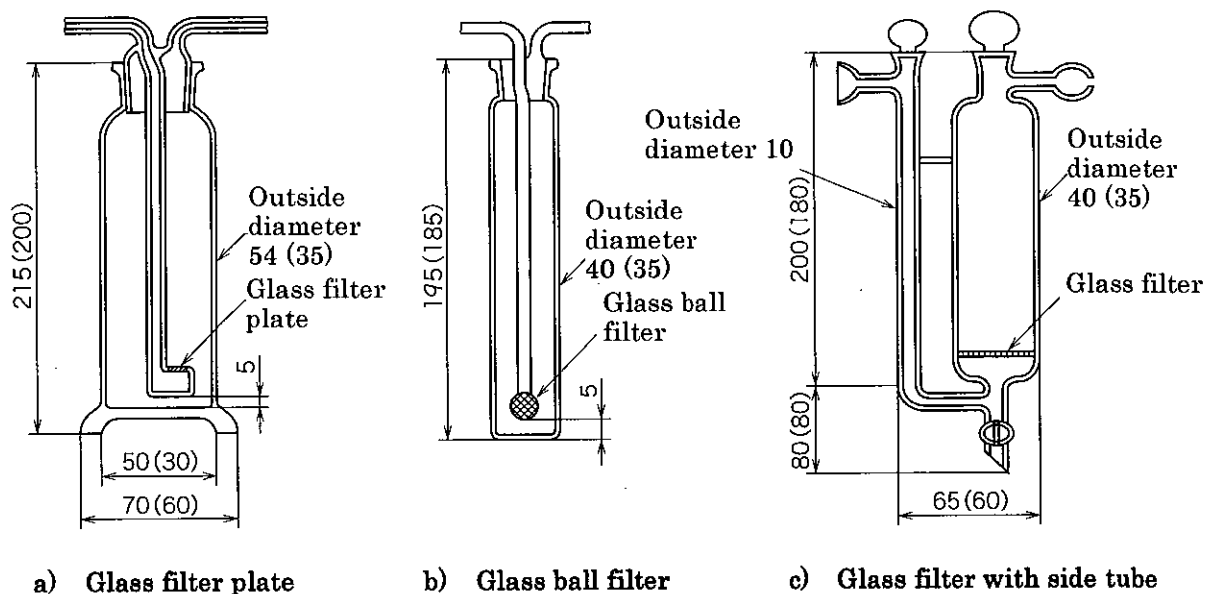
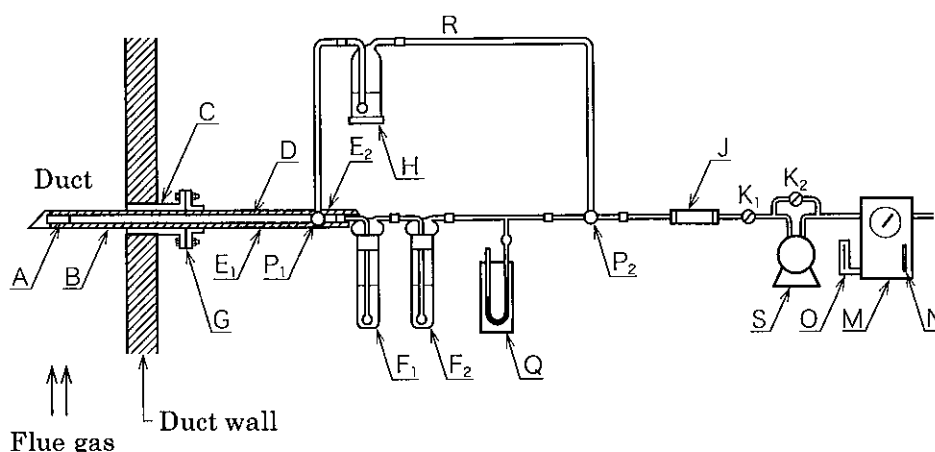


Figure 1 Example of absorbing bottle (100 ml and 250 ml)

For the pore size of glass filter plate, glass ball filter and glass filter, the pore size symbol G1 or G2 specified in JIS R 3503 shall be used.



- | | |
|--|---|
| A : filtering medium | K ₁ , K ₂ : flow rate control cock |
| B : sampling tube of sample gas | M : wet gas meter or dry gas meter |
| C : inlet | N : thermometer |
| D : thermometer | O : manometer |
| E ₁ , E ₂ : heater, ribbon heater | P ₁ , P ₂ : three-way cock for flow path switch |
| F ₁ , F ₂ : absorbing bottle (250 ml or 100 ml
in capacity) | Q : mercury manometer |
| G : flange | R : bypass |
| H : washing bottle (put 50 ml of absorp-
tion solution) | S : suction pump |
| J : drying tube | |

Figure 2 Example of sampling apparatus of sample gas

5.4 Sampling procedure

The sampling procedure shall be as follows.

The symbols of apparatus adopted here shall be as given in figure 2.

- a) The absorbing bottle and the liquid amount for absorption shall be either of the following.
 - 1) **Case of ion chromatography** Put 25 ml of absorption solution of 5.2.2 b) into each of the absorbing bottles (100 ml in capacity) (F₁ and F₂). When the concentration of sulfur oxides is expected to exceed 58 vol ppm, put 50 ml of absorption solution of 5.2.2 a) into each of the absorbing bottles (250 ml in capacity) (F₁ and F₂).
 - 2) **Case of precipitation titrimetry (arsenazo III method)** Put 50 ml of absorption solution of 5.2.2 a) into each of the absorbing bottles (250 ml in capacity) (F₁ and F₂).
- b) Turn the three-way cocks for flow path switch (P₁ and P₂) to bypass (R) side, adjust the flow rate 1 L/min to 2 L/min, start the suction pump (S), and replace inside air between the sampling tube of sample gas (B) and the three-way cock for flow path switch (P₁) by the sample gas.

NOTE 1 The bypass (R) between P₁ and P₂ in figure 2 may be omitted, provided that the distance between the sampling tube of sample gas (B) and the absorbing bottle is short and the flow rate can be adjusted to a definite amount.

NOTE 2 When it can be confirmed that the leak does not occur in the sampling apparatus by other technique, a mercury manometer (Q) in figure 2 may be omitted.

- c) After stopping the suction pump (S), close the three-way cocks for flow path switch (P₁ and P₂) toward the absorbing bottles' (F₁ and F₂) side. Then, read the indication (V₁) of a gas meter (M) to the nearest 0.01 L.
- d) Start the suction pump (S) and pass the sample gas through the absorbing bottles (F₁ and F₂). At this time, adjust a flow rate to about 1 L/min by adjusting the flow rate control cocks (K₁ and K₂).

If there is a possibility that the absorption solution is warmed, place the absorbing bottles in a cooling tank. In this case, the absorbing bottle shaped as figure 1 a) should be used. When the adsorbing bottle of 250 ml in capacity is used in ion chromatography and when it has been known that sulfur oxides will be completely absorbed in the absorption solution in precipitation titrimetry, the flow rate may be increased up to 2 L/min.

- e) Measure the temperature and the gauge pressure of the gas meter with the thermometer (N) and the manometer (O) of the gas meter (M). Also, measure the atmospheric pressure.
- f) After sampling about 20 L of sample gas, stop the suction pump (S), close the three-way cocks for flow path switch (P₁ and P₂), and read the indication (V₂) of a gas meter (M) to the nearest 0.01 L.

This amount may be suitably increased/decreased according to the sulfur oxides concentration.

- g) If necessary, measure the moisture in the sample gas in accordance with clause 6 of JIS Z 8808.

5.5 Sampling volume of sample gas

According to the following formulae, the sampling volume of sample gas under the standard conditions [273.15 K (0 °C), 101.32 kPa] shall be converted into the dry-base gas volume (V_{SD}) or the wet-base gas volume (V_{SW}).

5.5.1 When obtaining by dry-base gas amount

- a) When using a wet-base gas meter,

$$V_{SD} = V \times \frac{273.15}{273.15 + t} \times \frac{P_a + P_m - P_v}{101.32} + 22.41(a + b) \dots\dots\dots(1)$$

- b) When using a dry-base gas meter,

$$V_{SD} = V \times \frac{273.15}{273.15 + t} \times \frac{P_a + P_m}{101.32} + 22.41(a + b) \dots\dots\dots(2)$$

5.5.2 When obtaining by wet-base gas amount

a) When using a wet-base gas meter,

$$V_{\text{SW}} = V \times \frac{273.15}{273.15 + t} \times \frac{P_a + P_m - P_v}{101.32} + 22.41(a + b + c) \quad \text{.....(3)}$$

b) When using a dry-base gas meter,

$$V_{\text{SW}} = V \times \frac{273.15}{273.15 + t} \times \frac{P_a + P_m}{101.32} + 22.41(a + b + c) \quad \text{.....(4)}$$

where, V_{SD} : dry-base gas volume (L)

V_{SW} : wet-base gas volume (L)

V : gas volume measured by gas meter (L)
[equals to $V_2 - V_1$ at procedures of 5.4 c) and f)]

t : temperature at gas meter ($^{\circ}\text{C}$)

P_a : atmospheric pressure (kPa)

$P_m^{2)}$: gauge pressure at gas meter (kPa)

P_v : saturated water vapour pressure at $t^{\circ}\text{C}$ (kPa)

$a^{2)}$: analyte gas collected in absorption solution (mol)

$b^{2)}$: gases other than analyte gas collected in absorption solution (mol)

$c^{2)}$: moisture amount obtained in clause 6 of JIS Z 8808 (mol)

22.41 : volume of 1 mol gas under standard condition (L)

Note ²⁾ In most cases, it can be neglected.

P_v shall be as given in table 2.

Table 2 Saturated vapour pressure of water

Unit : Pa

t/°C	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.	611.21	615.67	620.15	624.67	629.21	633.78	638.38	643.01	647.67	652.36
1.	657.08	661.83	666.61	671.42	676.26	681.14	686.04	690.98	695.94	700.94
2.	705.97	711.03	716.13	721.26	726.41	731.61	736.83	742.09	747.38	752.70
3.	758.06	763.45	768.88	774.34	779.83	785.36	790.92	796.52	802.15	807.82
4.	813.52	819.26	825.03	830.84	836.69	842.57	848.49	854.45	860.44	866.47
5.	872.54	878.64	884.79	890.97	897.19	903.44	909.74	916.07	922.45	928.86
6.	935.31	941.80	948.34	954.91	961.52	968.17	974.86	981.60	988.37	995.19
7.	1 002.0	1 008.9	1 015.9	1 022.9	1 029.9	1 037.0	1 044.1	1 051.2	1 058.4	1 065.7
8.	1 072.9	1 080.3	1 087.6	1 095.1	1 102.5	1 110.0	1 117.6	1 125.2	1 132.8	1 140.5
9.	1 148.2	1 156.0	1 163.8	1 171.7	1 179.6	1 187.6	1 195.6	1 203.7	1 211.8	1 219.9
10.	1 228.1	1 236.4	1 244.7	1 253.0	1 261.4	1 269.9	1 278.4	1 286.9	1 295.5	1 304.2
11.	1 312.9	1 321.7	1 330.5	1 339.3	1 348.2	1 357.2	1 366.2	1 375.3	1 384.4	1 393.5
12.	1 402.8	1 412.1	1 421.4	1 430.8	1 440.2	1 449.7	1 459.3	1 468.9	1 478.5	1 488.2
13.	1 498.0	1 507.8	1 517.7	1 527.7	1 537.7	1 547.7	1 557.9	1 568.0	1 578.3	1 588.6
14.	1 598.9	1 609.3	1 619.8	1 630.3	1 640.9	1 651.6	1 662.3	1 673.0	1 683.9	1 694.8
15.	1 705.7	1 716.7	1 727.8	1 739.0	1 750.2	1 761.4	1 772.8	1 784.2	1 795.6	1 807.1
16.	1 818.7	1 830.4	1 842.1	1 853.9	1 865.8	1 877.7	1 889.7	1 901.7	1 913.8	1 926.0
17.	1 938.3	1 950.6	1 963.0	1 975.5	1 988.0	2 000.6	2 013.3	2 026.0	2 038.8	2 051.7
18.	2 064.7	2 077.7	2 090.8	2 104.0	2 117.2	2 130.5	2 143.9	2 157.4	2 170.9	2 184.5
19.	2 198.2	2 212.0	2 225.8	2 239.7	2 253.7	2 267.8	2 281.9	2 296.1	2 310.4	2 324.8
20.	2 339.2	2 353.8	2 368.4	2 383.1	2 397.8	2 412.7	2 427.6	2 442.6	2 457.7	2 472.9
21.	2 488.2	2 503.5	2 518.9	2 534.4	2 550.0	2 565.7	2 581.4	2 597.3	2 613.2	2 629.2
22.	2 645.3	2 661.5	2 677.7	2 694.1	2 710.5	2 727.1	2 743.7	2 760.4	2 777.2	2 794.1
23.	2 811.0	2 828.1	2 845.2	2 862.5	2 879.8	2 897.2	2 914.8	2 932.4	2 950.1	2 967.9
24.	2 985.8	3 003.7	3 021.8	3 040.0	3 058.3	3 076.6	3 095.1	3 113.6	3 132.3	3 151.1
25.	3 169.9	3 188.9	3 207.9	3 227.0	3 246.3	3 265.6	3 285.1	3 304.6	3 324.3	3 344.0
26.	3 363.9	3 383.8	3 403.9	3 424.0	3 444.3	3 464.7	3 485.2	3 505.7	3 526.4	3 547.2
27.	3 568.1	3 589.1	3 610.2	3 631.5	3 652.8	3 674.2	3 695.8	3 717.4	3 739.2	3 761.1
28.	3 783.1	3 805.2	3 827.4	3 849.7	3 872.2	3 894.7	3 917.4	3 940.2	3 963.1	3 986.1
29.	4 009.2	4 032.5	4 055.8	4 079.3	4 102.9	4 126.6	4 150.5	4 174.4	4 198.5	4 222.7
30.	4 247.0	4 271.5	4 296.0	4 320.7	4 345.5	4 370.5	4 395.5	4 420.7	4 446.0	4 471.5
31.	4 497.0	4 522.7	4 548.5	4 574.5	4 600.5	4 626.7	4 653.1	4 679.5	4 706.1	4 732.8
32.	4 759.7	4 786.7	4 813.8	4 841.0	4 868.4	4 895.9	4 923.6	4 951.4	4 979.3	5 007.4
33.	5 035.6	5 063.9	5 092.4	5 121.0	5 149.7	5 178.6	5 207.7	5 236.8	5 266.2	5 295.6
34.	5 325.2	5 355.0	5 384.8	5 414.9	5 445.1	5 475.4	5 505.9	5 536.5	5 567.2	5 598.1
35.	5 629.2	5 660.4	5 691.8	5 723.3	5 754.9	5 786.8	5 818.7	5 850.8	5 883.1	5 915.5
36.	5 948.1	5 980.8	6 013.7	6 046.8	6 080.0	6 113.3	6 146.9	6 180.5	6 214.4	6 248.4
37.	6 282.5	6 316.9	6 351.3	6 386.0	6 420.8	6 455.8	6 490.9	6 526.2	6 561.7	6 597.3
38.	6 633.1	6 669.1	6 705.2	6 741.5	6 778.0	6 814.7	6 851.5	6 888.5	6 925.6	6 963.0
39.	7 000.5	7 038.2	7 076.0	7 114.1	7 152.3	7 190.7	7 229.2	7 268.0	7 306.9	7 346.0
40.	7 385.3	7 424.8	7 464.4	7 504.2	7 544.3	7 584.5	7 624.8	7 665.4	7 706.2	7 747.1

Table 2 (continued)

Unit : Pa

$t/^{\circ}\text{C}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
41.	7 788.2	7 829.6	7 871.1	7 912.8	7 954.6	7 996.7	8 039.0	8 081.5	8 124.1	8 167.0
42.	8 210.0	8 253.2	8 296.7	8 340.3	8 384.1	8 428.2	8 472.4	8 516.8	8 561.5	8 606.3
43.	8 651.3	8 696.5	8 742.0	8 787.6	8 833.5	8 879.5	8 925.8	8 972.3	9 018.9	9 065.8
44.	9 112.9	9 160.2	9 207.7	9 255.5	9 303.4	9 351.6	9 399.9	9 448.5	9 497.3	9 546.3
45.	9 595.6	9 645.0	9 694.7	9 744.6	9 794.7	9 845.0	9 895.6	9 946.4	9 997.4	10 049.
46.	10 100.	10 152.	10 204.	10 256.	10 308.	10 361.	10 414.	10 467.	10 520.	10 573.
47.	10 627.	10 681.	10 735.	10 790.	10 845.	10 899.	10 955.	11 010.	11 066.	11 122.
48.	11 178.	11 234.	11 291.	11 348.	11 405.	11 462.	11 520.	11 578.	11 636.	11 694.
49.	11 753.	11 812.	11 871.	11 930.	11 990.	12 049.	12 110.	12 170.	12 231.	12 292.
50.	12 353.	12 414.	12 476.	12 538.	12 600.	12 663.	12 725.	12 788.	12 852.	12 915.
51.	12 979.	13 043.	13 107.	13 172.	13 237.	13 302.	13 368.	13 433.	13 499.	13 566.
52.	13 632.	13 699.	13 766.	13 833.	13 901.	13 969.	14 037.	14 106.	14 175.	14 244.
53.	14 313.	14 383.	14 453.	14 523.	14 594.	14 665.	14 736.	14 807.	14 879.	14 951.
54.	15 023.	15 096.	15 169.	15 242.	15 316.	15 389.	15 464.	15 538.	15 613.	15 688.
55.	15 763.	15 839.	15 915.	15 991.	16 068.	16 145.	16 222.	16 299.	16 377.	16 455.
56.	16 534.	16 613.	16 692.	16 771.	16 851.	16 931.	17 012.	17 093.	17 174.	17 255.
57.	17 337.	17 419.	17 501.	17 584.	17 667.	17 750.	17 834.	17 918.	18 003.	18 087.
58.	18 173.	18 258.	18 344.	18 430.	18 516.	18 603.	18 690.	18 778.	18 866.	18 954.
59.	19 043.	19 131.	19 221.	19 310.	19 400.	19 491.	19 581.	19 672.	19 764.	19 856.
60.	19 948.	20 040.	20 133.	20 226.	20 320.	20 414.	20 508.	20 603.	20 698.	20 793.
61.	20 889.	20 985.	21 082.	21 179.	21 276.	21 374.	21 472.	21 571.	21 669.	21 769.
62.	21 868.	21 968.	22 069.	22 170.	22 271.	22 372.	22 474.	22 577.	22 679.	22 783.
63.	22 886.	22 990.	23 094.	23 199.	23 304.	23 410.	23 516.	23 622.	23 729.	23 836.
64.	23 944.	24 052.	24 160.	24 269.	24 379.	24 488.	24 598.	24 709.	24 820.	24 931.
65.	25 043.	25 155.	25 268.	25 381.	25 494.	25 608.	25 723.	25 837.	25 953.	26 068.
66.	26 184.	26 301.	26 418.	26 535.	26 653.	26 772.	26 890.	27 010.	27 129.	27 249.
67.	27 370.	27 491.	27 612.	27 734.	27 857.	27 979.	28 103.	28 226.	28 351.	28 475.
68.	28 600.	28 726.	28 852.	28 979.	29 106.	29 233.	29 361.	29 489.	29 618.	29 748.
69.	29 877.	30 008.	30 138.	30 270.	30 402.	30 534.	30 667.	30 800.	30 933.	31 068.
70.	31 202.	31 338.	31 473.	31 609.	31 746.	31 883.	32 021.	32 159.	32 298.	32 437.
71.	32 577.	32 717.	32 858.	32 999.	33 140.	33 283.	33 425.	33 569.	33 713.	33 857.
72.	34 002.	34 147.	34 293.	34 439.	34 586.	34 734.	34 882.	35 030.	35 179.	35 329.
73.	35 479.	35 630.	35 781.	35 933.	36 085.	36 238.	36 391.	36 545.	36 700.	36 855.
74.	37 010.	37 166.	37 323.	37 480.	37 638.	37 796.	37 955.	38 115.	38 275.	38 436.
75.	38 597.	38 758.	38 921.	39 084.	39 247.	39 411.	39 576.	39 741.	39 907.	40 073.
76.	40 240.	40 408.	40 576.	40 744.	40 914.	41 084.	41 254.	41 425.	41 597.	41 769.
77.	41 942.	42 116.	42 290.	42 464.	42 640.	42 815.	42 992.	43 169.	43 347.	43 525.
78.	43 704.	43 884.	44 064.	44 245.	44 426.	44 608.	44 791.	44 974.	45 158.	45 343.
79.	45 528.	45 714.	45 900.	46 088.	46 275.	46 464.	46 653.	46 843.	47 033.	47 224.
80.	47 416.	47 608.	47 801.	47 994.	48 189.	48 384.	48 579.	48 776.	48 972.	49 170.

Table 2 (concluded)

Unit : Pa

$t/^\circ\text{C}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
81.	49 368.	49 567.	49 767.	49 967.	50 168.	50 370.	50 572.	50 775.	50 979.	51 183.
82.	51 388.	51 594.	51 800.	52 007.	52 215.	52 424.	52 633.	52 843.	53 053.	53 265.
83.	53 477.	53 689.	53 903.	54 117.	54 332.	54 547.	54 764.	54 981.	55 198.	55 417.
84.	55 636.	55 856.	56 076.	56 298.	56 520.	56 743.	56 966.	57 190.	57 415.	57 641.
85.	57 868.	58 095.	58 323.	58 552.	58 781.	59 011.	59 242.	59 474.	59 707.	59 940.
86.	60 174.	60 409.	60 644.	60 881.	61 118.	61 356.	61 594.	61 834.	62 074.	62 315.
87.	62 557.	62 799.	63 042.	63 286.	63 531.	63 777.	64 024.	64 271.	64 519.	64 768.
88.	65 017.	65 268.	65 519.	65 771.	66 024.	66 278.	66 532.	66 788.	67 044.	67 301.
89.	67 559.	67 817.	68 077.	68 337.	68 598.	68 860.	69 123.	69 386.	69 651.	69 916.
90.	70 182.	70 449.	70 717.	70 986.	71 255.	71 526.	71 797.	72 069.	72 342.	72 616.
91.	72 890.	73 166.	73 442.	73 719.	73 998.	74 277.	74 556.	74 837.	75 119.	75 401.
92.	75 685.	75 969.	76 254.	76 540.	76 827.	77 115.	77 404.	77 693.	77 984.	78 276.
93.	78 568.	78 861.	79 155.	79 450.	79 746.	80 043.	80 341.	80 640.	80 940.	81 240.
94.	81 542.	81 844.	82 148.	82 452.	82 757.	83 064.	83 371.	83 679.	83 988.	84 298.
95.	84 609.	84 921.	85 234.	85 547.	85 862.	86 178.	86 495.	86 812.	87 131.	87 451.
96.	87 771.	88 093.	88 415.	88 739.	89 063.	89 389.	89 715.	90 043.	90 371.	90 701.
97.	91 031.	91 362.	91 695.	92 028.	92 363.	92 698.	93 035.	93 372.	93 711.	94 050.
98.	94 391.	94 732.	95 075.	95 418.	95 763.	96 109.	96 455.	96 803.	97 152.	97 502.
99.	97 853.	98 204.	98 557.	98 911.	99 266.	99 623.	99 980.	100 338.	100 697.	101 058.
100.	101 419.	101 782.	102 145.	102 510.	102 875.	103 242.	103 610.	103 979.	104 349.	104 720.

NOTE : It is based on SONNTAG (1990). A temperature scale is ITS-90.

6 Preparation of sample solution for analysis

Preparation shall be carried out according to any one of the following.

6.1 Case of ion chromatography

- After the completion of the procedure of 5.4, transfer the content solution in the absorbing bottles (F_1 and F_2) into a beaker of 300 ml. Besides, wash the absorbing bottles, etc. with water and transfer the washings into the beaker.
- When an absorbing bottle of 100 ml in capacity is used, transfer the content solution in the beaker into a 100 ml volumetric flask by washing with water and add water up to the marked line. Take this as the sample solution for analysis.
- When an absorbing bottle of 250 ml in capacity is used, transfer the contents solution in the beaker into a 250 ml volumetric flask by washing with water and add water up to the marked line. Take this as the sample solution for analysis. In this case, to lessen the influence on a separation column, it shall be diluted by 10 times or more.
- If solid precipitate is found in the sample solutions for analysis of b) and c), eliminate it by filtering through the filter with 0.45 μm pore diameter or less.

6.2 Case of precipitation titrimetry (arsenazo III method)

- a) After the completion of the procedure of 5.4, carry out the procedure of 6.1 a).
- b) Transfer the content solution in the beaker into a 250 ml volumetric flask by washing with water and add water up to the marked line. Take this as the sample solution for analysis.

The measurement of this solution may be carried out by the ion chromatography. But in this case, to lessen the influence on a separation column, it shall be diluted by 10 times or more.

7 Determination by chemical analysis

7.1 Ion chromatography

7.1.1 Applicable condition

This method is affected by the high concentrated coexistence of reductive gas such as sulfides in the sample gas, therefore this can be applicable when such influence can be neglected or eliminated.

7.1.2 Reagents and preparation of reagent solutions

- a) **Reagents** Reagents shall be as follows.

- 1) **Sulfuric acid** That specified in JIS K 8951.
- 2) **Boric acid** That specified in JIS K 8863.
- 3) **Sodium hydrogen carbonate** That specified in JIS K 8622.
- 4) **Sodium carbonate** That specified in JIS K 8625.
- 5) **Sodium tetraborate decahydrate** That specified in JIS K 8866.
- 6) **Sodium sulfate** That specified in JIS K 8987.
- 7) **Potassium sulfate** That specified in JIS K 8962.
- 8) **Potassium gluconate**
- 9) **p-hydroxybenzoic acid**
- 10) **2-Bis(2-hydroxyethyl)amino-2-hydroxymethyl-1,3-propanediol(Bis-Tris)** That specified in JIS K 9808.
- 11) **Phthalic acid**
- 12) **2-amino-2-hydroxymethyl-1, 3-propanediol [tris (hydroxymethyl) aminomethane]** That specified in JIS K 9704.
- 13) **Acetonitrile** That specified in JIS K 8032.
- 14) **Glycerol** That specified in JIS K 8295.
- 15) **Sulfate ion reference solution (SO_4^{2-} : 1 000 mg/L)**

- b) **Preparation of reagent solutions**

- 1) **Eluant** The eluant differs depending on the type of apparatus and the type of separation column to be used, therefore that capable of separating the target ion by not less than 1.3 in resolution (R) respectively shall be used³⁾. The confirmation of resolution shall be in accordance with clause 10 of JIS K 0127.

NOTE : Examples of eluant are as follows.

Moreover, if it is confirmed that sulfate ions can be measured quantitatively, other eluant than that indicated herein may be used according to the characteristic of separation column.

- **Hydrogen carbonate-carbonate solution (1)** Dissolve 0.025 g (0.3 mmol) of sodium hydrogen carbonate and 0.286 g (2.7 mmol) of sodium carbonate in water, and transfer it into a 1 000 ml volumetric flask by washing with water and add water up to the marked line.
- **Hydrogen carbonate-carbonate solution (2)** Dissolve 0.143 g (1.7 mmol) of sodium hydrogen carbonate and 0.191 g (1.8 mmol) of sodium carbonate in water, and transfer it into a 1 000 ml volumetric flask by washing with water and add water up to the marked line.
- **Gluconate-tetra borate-boric acid solution** Dissolve 0.305 g (1.3 mmol) of potassium gluconate, 0.496 g (1.3 mmol) of sodium tetra borate decahydrate, 1.855 g (30 mmol) of boric acid, 100 ml of acetonitrile and 5 ml of glycerine in water, and transfer it into a 1 000 ml volumetric flask by washing with water and add water up to the marked line.
- ***p*-hydroxybenzoic acid-2-[bis(2-hydroxyethyl)aminol]-2-hydroxymethyl-1,3-propanediol solution** Dissolve 1.105 g (8.0 mmol) of *p*-hydroxybenzoic acid and 0.669 g (3.2 mmol) of 2-bis(2-hydroxyethyl) amino-2-hydroxymethyl-1,3-propanediol in water, and transfer it into a 1 000 ml volumetric flask by washing with water and add water up to the marked line.
- **Phthalic acid-2-amino-2-hydroxymethyl-1,3-propanediol solution** Dissolve 0.415 g (2.5 mmol) of phthalic acid and 0.290 g (2.4 mmol) of 2-amino-2-hydroxymethyl-1,3-propanediol, or 0.382 g (2.3 mmol) of phthalic acid and 0.303 g (2.5 mmol) of 2-amino-2-hydroxymethyl-1,3-propanediol in water, and transfer it into a 1 000 ml volumetric flask by washing with water and add water up to the marked line.

- 2) **Regeneration solution (removal solution)** This solution is used for regeneration or maintaining continuously the function of a suppressor, and used for regeneration with electrical or chemical methods. The optimum solution shall be used for the type of apparatus and suppressor.

NOTE : Examples of regeneration solution and regenerating material are as follows.

- **Water** That in 3 e) is decomposed by an electric current and used for the suppressor in the system producing regeneration

solution.

- **Eluant** That passed through a detector is decomposed by an electric current and used for the suppressor in the system producing regeneration solution.
- **Sulfuric acid (12.5 mmol/L)** 12.5 ml of sulfuric acid (1 mol/L) is made 1 L with water. This is taken as the regeneration solution.
- **Ion exchange resin** Cation exchanger is mixed in eluant.

3) **Sulfate ion reference solution (SO_4^{2-} : 1 mg/ml)** Sulfate ion reference solution (SO_4^{2-} : 1 000 mg/L) of a) 15), or sodium sulfate or potassium sulfate is dissolved in water and prepared. In the case of sodium sulfate, prepare it as follows: heat it at about 110 °C for about 2 h, allow it to cool in a desiccator, weigh its 1.479 g, dissolve it in water, transfer it into a 1 000 ml volumetric flask while rinsing with water, and add water up to the marked line. In the case of potassium sulfate, prepare it as follows: ignite it at about 750 °C for about 15 min, weigh its 1.814 g after cooling it, carry out similarly above, and make the total 1 000 ml.⁴⁾

4) **Sulfate ion reference solution (SO_4^{2-} : 0.2 mg/ml)** Take accurately 50 ml of sulfate ion reference solution (SO_4^{2-} : 1 mg/ml) prepared in 3) into a 250 ml volumetric flask, and add water up to the marked line. Prepare it at every service.

Notes ³⁾ Choose it preferably making reference to the operating manuals of apparatus and column.

⁴⁾ Care shall be taken into moisture while weighing. Sodium sulfate is highly hygroscopic, therefore it shall be treated as follows : take it in a weighing bottle by a little more than needed, dry it, allow it to cool, weigh it accurately, then dissolve it in water. Concentration shall be calculated on the base of the taken amount.

7.1.3 Implements and apparatus

The implements and apparatus shall be as follows.

- a) **Sample introducing device** That of automatic type by which a definite amount of the sample solution for analysis is introduced into the apparatus with superior reproducibility, or that of manual type by which the sample solution for analysis is introduced into a sample measuring tube (a definite amount of 10 µl to 250 µl) built-in the apparatus by using a syringe from 1 ml to 10 ml.
- b) **Ion chromatograph** For the ion chromatograph, either of the suppressor system or the non-suppressor system may be used.
 - 1) **Separation column** Tube of 2 mm to 8 mm in inside diameter and 30 mm to 300 mm in length made of inactive synthetic resin or metal, which is packed with an ion exchanger and capable of separating the target ion from the adjacent ion by not less than 1.3 in resolution.
 - 2) **Pre-column** Guard column which is used for concentration, pre-separation and foreign matter elimination set before the separation column, if necessary. It shall

be the tube of 2 mm to 6 mm in inside diameter and 5 mm to 50 mm in length made of inactive synthetic resin or metal, which is packed with anion exchanger of the same type as that of the separation column.

- 3) **Suppressor** ⁵⁾ Implement which decreases the electric conductivity by changing the eluant electrically or chemically, in order to measure ion species in the eluant with high sensitivity by using the electric conductivity detector. There are the dialysis type, the column removal type and the suspension resin adsorption type.
- 4) **Detector** Electric conductivity detector.
- 5) **Recording part** In accordance with 4.2 f) of JIS K 0127.

Note ⁵⁾ Although the type equipped with suppressor is specified in ISO 11632, either can be serviceable in this Standard.

7.1.4 Procedure for determination

The procedure for determination shall be as follows.

- a) Set an ion chromatograph to be ready for measurement, and flow eluant through a separation column at a constant flow rate (e.g., 1 ml/min to 2 ml/min). For the apparatus with suppressor, flow eluant through inside of the separation column and the suppressor. Also, flow the regeneration solution through outside of the suppressor at a constant flow rate.
- b) Introduce a definite amount (10 μ l to 250 μ l) of sample solution for analysis prepared in 6.1 using the sample introducing device into the ion chromatograph, and record the chromatogram.

When the sample of specially low concentration is measured, instead of the sample measuring tube in the apparatus, the concentrating column should be preferably used, with which anion exchanger of the same type as that in the separation column is packed.

- c) Obtain the peak area or the peak height from the peak corresponding to the sulfate ion on chromatogram.
- d) Obtain the sulfate ion concentration (mg/ml) with the working curve prepared in 7.1.5.
- e) Take 50 ml of absorption solution into a 100 ml volumetric flask, add water up to the marked line, use the same amount as introduced in b), carry out the procedures in accordance with b) and c), and obtain the blank test value (mg/ml) of sulfate ion.

When the absorbing bottle of 250 ml in capacity is used, take 100 ml of absorption solution into a 250 ml volumetric flask, add water up to the marked line, and carry out the same procedures.

7.1.5 Preparation of working curve

The preparation of the working curve shall be as follows.

- a) Take 0.2 ml to 25.0 ml of the sulfate ion reference solution (SO_4^{2-} : 0.2 mg/ml) stepwise into several 100 ml volumetric flasks, add water up to the marked line

respectively, and obtain each concentration. For the sulfate ion reference solution, take several points within either range of 0.2 ml to 5.0 ml or 2.0 ml to 25.0 ml corresponding to the expected concentration of sample.

In the case of using the absorbing bottle of 250 ml in capacity, take 1.0 ml to 25.0 ml of sulfate ion reference solution (SO_4^{2-} : 1 mg/ml) stepwise into several 250 ml volumetric flasks, add water up to the marked line, and carry out the same procedure and prepare the working curve.

- b) Carry out the procedures of b) and c) of 7.1.4, and obtain the peak area or the peak height which corresponds to each sulfate ion concentration.
- c) As the blank test, carry out the procedures of b) and c) of 7.1.4 for water, obtain the peak area or the peak height which corresponds to each sulfate ion concentration.
- d) Prepare the relation curve between the sulfate ion concentration and the peak area or the peak height which has been corrected by the blank test value.

Prepare the working curve on every measurement of the sample.

7.1.6 Calculation

The sulfur oxides concentration in the sample gas shall be calculated according to formula (5) to formula (7).

$$C_V = \frac{0.233 \times (a - b) \times v}{V_S} \times 1\,000 \quad \text{.....(5)}$$

$$C_W = \frac{0.667 \times (a - b) \times v}{V_S} \times 1\,000 \quad \text{.....(6)}$$

$$C_W = C_V \times 2.86 \quad \text{.....(7)}$$

- where,
- C_V : volume fraction of sulfur oxides in sample gas (vol ppm)
 - C_W : mass concentration of sulfur oxides in sample gas expressed as sulfur dioxide (mg/m^3)
 - a : sulfate ion concentration obtained in 7.1.4 d) (mg/ml)
 - b : sulfate ion concentration obtained by blank test in 7.1.4 e) (mg/ml)
 - v : volumetric flask capacity (100 for 100 ml and 250 for 250 ml) (ml)
 - V_S : sample gas volume sampled under standard condition calculated in 5.5 (L)
(V_{SD} for dry-base gas volume and V_{SW} for wet-base gas volume)
 - 0.233 : volume of sulfur oxides ($\text{SO}_2 + \text{SO}_3$) equivalent to 1 mg of sulfate ion concentration (SO_4^{2-}) (ml) (standard condition)
 - 0.667 : mass of sulfur dioxide (SO_2) equivalent to 1 mg of sulfate ion concentration (SO_4^{2-}) (mg)
 - 2.86 : mass concentration of sulfur dioxide (SO_2) equivalent to 1 vol ppm of sulfur oxides (mg/m^3) (according to 64.06/22.41)

NOTE : Hydrogen chloride and nitrogen oxides can be measured simultaneously with sulfur oxides according to this method. The procedures in this case are in accordance with Annex JC and Annex JD.

7.2 Precipitation titrimetry (arsenazo III method)

7.2.1 Reagents and preparation of reagent solutions

a) Reagents

- 1) Sulfuric acid That specified in JIS K 8951.
- 2) Acetic acid That specified in JIS K 8355.
- 3) Lead acetate (II) trihydrate That specified in JIS K 8374.
- 4) Barium acetate That specified in JIS K 8376.
- 5) Bromophenol blue That specified in JIS K 8844.
- 6) Arsenazo III That specified in JIS K 9524.
- 7) Ethanol (95) That specified in JIS K 8102.
- 8) 2-Propanol That specified in JIS K 8839.
- 9) Sodium carbonate The reference material for volumetric analysis specified in JIS K 8005.

b) Preparation of reagent solutions

- 1) Bromophenol blue solution Dissolve 0.1 g of bromophenol blue in 20 ml of ethanol (95), and add water to make the total 100 ml.
- 2) Arsenazo III solution Weigh 0.2 g of arsenazo III, add 100 ml of water to dissolve it by shaking sufficiently, filtrate it, and store it in a brown bottle. Do not use the solution elapsed 1 month or longer after preparation.
- 3) 2 mmol/L sulfuric acid Add 3 ml of sulfuric acid into about 1 L of water to prepare 0.05 mol/L sulfuric acid, and use it diluted with water by accurately 25 times.

Standardize 0.05 mol/L sulfuric acid as follows.

- Take 1 g to 1.5 g of sodium carbonate of reference material for volumetric analysis into a platinum crucible, heat it at 600 °C for about 60 min, allow it to cool in a desiccator, and weigh its mass to the nearest 0.1 mg. Dissolve it in a small amount of water, transfer it into a 250 ml volumetric flask while rinsing with water, and add water up to the marked line.
- Take accurately 25 ml out of this solution into a 200 ml conical beaker, add 2 or 3 drops of bromophenol blue solution, and titrate it with 0.05 mol/L sulfuric acid prepared here.
- When an end point comes near, boil it to expel carbon dioxide, cool it, and titrate on to find an end point where the colour of solution turns yellow from blue.
- Calculate the factor according to formula (8) ⁶⁾.

$$f' = \frac{m \times \frac{b}{100} \times \frac{25}{250}}{a' \times 0.005\,299} \dots\dots\dots(8)$$

where, f' : factor of 0.05 mol/L sulfuric acid
 m : amount of sodium carbonate taken (g)
 a' : amount of 0.05 mol/L sulfuric acid consumed in titration (ml)
 b : purity of sodium carbonate (%)
0.005 299 : amount of sodium carbonate equivalent to 1 ml of 0.05 mol/L sulfuric acid (g)

Note ⁶⁾ When making standardization, instead of sodium carbonate, 0.05 mol/L sodium hydroxide solution may be used for the reference.

- 4) **5 mmol/L barium acetate solution** Dissolve 1.1 g of barium acetate and 0.4 g of lead acetate (II) trihydrate in 200 ml of water and 3 ml of acetic acid, and add 2-propanol to make the total 1 L.

Standardize it as follows.

- Take accurately 10 ml out of 2 mmol/L sulfuric acid into a 200 ml Erlenmeyer flask, add 40 ml of 2-propanol, 1 ml of acetic acid and 4 to 6 drops of arsenazo III solution, and titrate it with 5 mmol/L barium acetate solution prepared here using a 5 ml microburette to find an end point where blue colour in solution lasts for 1 min.
- Calculate the factor according to formula (9).

$$f = \frac{10 \times f'}{a''} \times \frac{2}{5} \dots\dots\dots(9)$$

where, f : factor of 5 mmol/L barium acetate solution
 f' : factor of 2 mmol/L sulfuric acid obtained in 3)
 a'' : amount of 5 mmol/L barium acetate solution consumed in titration (ml)

It is preferable to employ photometric titration (in the vicinity of 600 nm wavelength) which can give clear end point.

7.2.2 Procedure for determination

The procedure for determination shall be as follows.

- a) Take 10 ml out of the sample solution for analysis prepared at 6.2 into a 200 ml Erlenmeyer flask.
- b) Add 40 ml of 2-propanol, 1 ml of acetic acid and 4 to 6 drops of arsenazo III solution, then titrate it with 5 mmol/L barium acetate solution using a 5 ml microburette to find an end point where blue colour of solution lasts for 1 min.

It is preferable to employ photometric titration (in the vicinity of 600 nm wavelength) which can give clear end point.

- c) Take 100 ml of absorption solution into a 250 ml volumetric flask, add water up to the marked line, take 10 ml from it into a 200 ml conical beaker, and titrate according to b) to find the blank test value.

Warning : Care shall be taken on handling because lead and arsenic are contained in this waste solution.

7.2.3 Calculation

Calculate the concentration of sulfur oxides in the sample gas according to formula (10) to formula (12).

$$C_V = \frac{0.112 \times (a - b) \times f \times \frac{250}{10}}{V_S} \times 1\,000 \dots\dots\dots(10)$$

$$C_W = \frac{0.320 \times (a - b) \times f \times \frac{250}{10}}{V_S} \times 1\,000 \dots\dots\dots(11)$$

$$C_W = C_V \times 2.86 \dots\dots\dots(12)$$

- where,
- C_V : volume fraction of sulfur oxides in sample gas (vol ppm)
 - C_W : mass concentration of sulfur oxides in sample gas expressed as sulfur dioxide (mg/m^3)
 - a : amount of 5 mmol/L barium acetate solution consumed in titration in 7.2.2 b) (ml)
 - b : amount of 5 mmol/L barium acetate solution consumed in blank test in 7.2.2 c) (ml)
 - f : factor of 5 mmol/L barium acetate solution
 - V_S : sample gas volume sampled under standard condition calculated in 5.5 (L)
(V_{SD} for dry-base gas volume and V_{SW} for wet-base gas volume)
 - 0.112 : volume of sulfur oxides ($\text{SO}_2 + \text{SO}_3$) equivalent to 1 ml of 5 mmol/L barium acetate solution (ml) (standard condition)
 - 0.320 : mass of sulfur dioxide (SO_2) equivalent to 1 ml of 5 mmol/L barium acetate solution (mg)
 - 2.86 : mass concentration of sulfur dioxide (SO_2) equivalent to 1 vol ppm of sulfur oxides (mg/m^3) (according to 64.06/22.41)

NOTE : This method is not influenced by coexistence of 0.5 mg or less of nitrogen oxides as NO_3^- in the solution taken out.

8 Determination of automated measuring method

The measurement according to the automated measuring method shall be carried out by the automated measuring instrument specified in JIS B 7981. The calibration shall be carried out as specified in JIS K 0055.

9 Method to obtain analysis value of flue gas

9.1 Case of chemical analysis

Carry out the analysis twice for the same sample solution prepared by every sampling, calculate their average, and round it off to two significant figures.

9.2 Case of automated analysis

Make the average of values measured in 1 h, read the value to the nearest 1/100 of the largest graduation in the measurable range, and round it off to two significant figures.

When adopting digital indication or print recording, read the figure which corresponds to it.

10 Record of analysis results

Items to be recorded as analysis results shall be as follows.

- a) Analysis value
- b) Type of analysis method (including type of chemical analysis method)
- c) Date and time of sampling
- d) Other necessary items

Annex A (normative)

Precipitation titrimetry (Thorin method) Stationary source emissions — Determination of the mass concentration of sulfur dioxide (SO₂) — Hydrogen peroxide/barium perchlorate/Thorin method

A.1 General

This Annex specifies a hydrogen peroxide/barium perchlorate/Thorin method for the determination of the mass concentration of sulfur dioxide emitted from combustion facilities and technical processes with negligible amounts of sulfur trioxide and sulfuric acid. It is applicable from a minimum of 30 mg/m³ sulfur dioxide by reference to sampling periods of normally 30 min.

At mass concentrations of sulfur dioxide greater than 2 000 mg/m³, the volume of the waste gas under investigation passed through the sampling train is 30 L. Substances, which, if contained in the waste gas under investigation and thus in the waste gas sample, are known to have an effect on the titration reading, are given in A.6.4. Information on performance characteristics is given in A.7.2.

At mass concentrations of sulfur dioxide less than 30 mg/m³, a sampling period greater than that specified in this Standard is used. All concentrations are based on dry gas at a temperature of 273.15 K and a pressure of 101.32 kPa.

A.2 Principle

Absorption of the sulfur dioxide (SO₂) present in the waste gas sample passing through a hydrogen peroxide solution within a specified period, resulting in the formation of sulfuric acid solution.

Adjustment of the pH of the sample solution to pH 3.5 with sodium hydroxide solution or perchloric acid solution as required. Determination of the mass concentration of sulfate ions present in the treated sample solution by titration with a barium perchlorate solution using Thorin as indicator and calculation of the mass concentration of sulfur dioxide.

A.3 Reagents and preparation of standard solutions

During the analysis use only reagents of recognized analytical grade and only water of at least Grade 3 purity according to ISO 3696.

WARNING: Use the reagents in accordance with the appropriate health and safety regulations.

A.3.1 2-Propanol Propanol [CH₃CH (OH) CH₃]

A.3.2 Absorption solution Place 100 ml of a 27 % (m/m) to 30 % (m/m) solution of

hydrogen peroxide (H_2O_2) into a 1 000 ml one-mark volumetric flask. Make up to the mark with water and mix well. Prepare this solution on the day of use.

A.3.3 Barium perchlorate standard volumetric solution $[\text{Ba}(\text{ClO}_4)_2]$ concentration = 0.005 mol/L.

Use a commercially available barium perchlorate solution of defined concentration or, if this is not possible, prepare for example as follows. Dissolve 1.7 g of anhydrous barium perchlorate $[\text{Ba}(\text{ClO}_4)_2]$ in about 200 ml of water in a 1 000 ml one-mark volumetric flask. Make up to the mark with 2-propanol (A.3.1) and mix well. Standardize this solution accurately by titration against a 0.005 mol/L standard volumetric sulfuric acid solution.

1 ml of exactly 0.005 mol/L barium perchlorate solution is equivalent to a mass of sulfur dioxide of 0.320 33 mg.

A.3.4 Potassium hydroxide standard volumetric solution (KOH) concentration = 0.1 mol/L.

A.3.5 Perchloric acid standard volumetric solution (HClO_4) concentration = 0.1 mol/L.

A.3.6 Thorin (4-[(2-arsenophenyl)-azol]-3-hydroxy-2, 7-naphthalene-disulfonic acid disodium salt) 2 g/L solution Dissolve 0.2 g of Thorin in small amount of water and transfer it into a 100 ml one-mark volumetric flask. Make up to the mark with water and mix well. Store this solution in a bottle made of fused silica or polyethylene.

A.4 Apparatus

A.4.1 Sampling equipment

A.4.1.1 Sampling probe Borosilicate glass or fused silica tube with a spherical ground joint at one end, of a suitable length to reach the representative measurement point(s) in the measurement plane of the waste gas flue and encircled by a heating jacket capable of producing a temperature of at least 200 °C.

NOTE 1 The heating jacket also serves as a protective tube to the gas sampling probe. The gas sampling probe should, therefore, always be used encircled by the heating jacket.

NOTE 2 A stop valve before the first absorber is necessary to prevent loss of agents when sampling in flues under suction conditions.

A.4.1.2 Particle filter Borosilicate glass or fused silica tube glass tube ends with spherical ground joints, packed in a progressive manner with chemically pure quartz wool. As an example, a particle filter that has proven suitable is shown in figure A.1.

A.4.1.3 Absorbers Absorption bottles of the Drechsel type, nominal capacity 100 ml or 250 ml as required (see table A.1) equipped with an absorption bottle insert having a sintered filter, the porosity of which shall be fine enough to enable an absorption efficiency of at least 95 % to be obtained. Sintered filters having pore diameters between 40 μm and 90 μm are suitable (see figure A.2).

The absorption efficiency of each individual absorber should be tested twice a year as specified in A.7.2.

NOTE : Impingers may be used if it is proven that absorption efficiency of at least 95 % can be achieved.

A.4.1.4 Heating bandage Capable of producing a temperature of at least 200 °C.

A.4.1.5 Voltage regulator

A.4.1.6 Trap Absorption bottle of the Drechsel type, equipped with an absorption bottle insert not having a sintered filter.

A.4.1.7 Sampling pump Capable of drawing waste gas at a volume flow rate within the range 0.02 m³/h to about 0.2 m³/h during the sampling period against a pressure of – 10 kPa to – 30 kPa.

A.4.1.8 Regulating valve Needle valve capable of adjusting a waste-gas flow rate within the range 0.02 m³/h to about 0.2 m³/h.

A.4.1.9 Gas metering device Wet-gas meter (or dry gas meter with a drying tube upstream) capable of use at a waste-gas volume flow rate within the range 0.02 m³/h to about 0.2 m³/h, limits of error of under 2 %, equipped with a thermometer.

Test the limits of error twice a year, to be within ± 2 %, by means of an appropriate calibrator whose traceability is secured.

A.4.1.10 Connecting tubing of different lengths and of different internal diameters, made of polyethylene, silicone rubber or polytetrafluoroethylene.

A.4.1.11 Thermometer measuring range – 5 °C to + 50 °C, limits of error $< \pm 0.2$ °C.

A.4.1.12 Barometer capable of measuring the atmospheric pressure present at the sampling location, limits of error approximately ± 1 % of the upper limit of measurement.

A.4.1.13 Stop watch

A.4.2 Direct reading pH meter preferably equipped with temperature compensation, measuring range 0 to 14, limits of error around pH 3.5 : < 0.2 .

Calibrate the direct reading pH meter in accordance with the manufacturer's instructions using an appropriate buffer solution, the pH of which shall be accurately known at a given temperature. After calibration, rinse the electrodes thoroughly before use.

A.5 Sampling

A.5.1 Depending on in which of the ranges given in table A.1 the expected mass concentration of sulfur dioxide lies, place, for example by pipette or dispenser, either 40 ml of the absorption solution (A.3.2) into each of the two 100 ml absorbers (A.4.1.3), or 80 ml of the absorption solution into each of the two 250 ml absorbers (A.4.1.3). Assemble the absorbers used and the apparatus specified in A.4.1.1 to A.4.1.2 and A.4.1.6 to A.4.1.11 to obtain a sampling train in accordance with the examples schematically rep-

resented in figure A.3. Use spherical ground glass joints upstream of the second absorber.

A.5.2 Plug the intake of the sampling probe (A.4.1.1), switch on the sampling pump (A.4.1.7) and leak test the sampling train following normal laboratory practice.

Carefully release the sampling probe intake plug and switch off the sampling pump.

A.5.3 Insert the sampling probe into the access hole in the wall of the waste gas flue and place the tip of the sampling probe at the representative measurement point (s) in the measurement plane of the waste gas flue. Fill the resultant space between the sampling probe and the access hole in the wall of the waste gas flue with an appropriate sealing material such that ambient air does not reach the measurement point (s) nor waste gas escape to any great degree.

A.5.4 Take necessary measures to stop ambient air entering the waste gas flue through the space between the sampling probe and the heating jacket, thus causing a local cooling of the sampling probe and altering the sample composition.

Wrap the particle filter (A.4.1.2) in the heating bandage (A.4.1.4). Switch on the heating system and adjust the voltage regulator (A.4.1.5) to give a temperature at each point within the heated zone at which, during the sampling period, the formation of condensate will not be possible.

NOTE : The temperature of the heating bandage may be checked with a thermocouple.

At the end of the heating up period (about half an hour) record the reading on the gas meter (A.4.1.9) and the time, start the sampling pump and adjust the regulating valve (A.4.1.8) to give the desired waste gas volume flow rate within the range of 0.03 m³/h to 0.2 m³/h.

A.5.5 Normally, the sampling period is 30 min. Record the reading on the gas meter thermometer (A.4.1.11), θ_i , and the reading on the barometer (A.4.1.12). The waste gas volume flow rate selected shall be approximately constant. The sample flow rate is derived from the information given on the sample volume of waste gas given in table A.1, and the normal sampling period (i.e. 30 min).

A.5.6 At the end of the sampling period, switch off the sampling pump, record the time and the reading on the gas meter. Remove the absorbers from the sampling train and transfer quantitatively both sample solutions into a sample bottle of suitable capacity.

Rinse the absorption bottle including the absorption bottle inserts with water and force the wash water through the sintered filters using a pressure bulb. Add the wash water to the combined sample solution in the sample bottle.

A.5.7 Adjust, using the direct reading pH meter (A.4.2), the pH of the combined sample solution to pH 3.5 by adding the appropriate volume of the standard volumetric 0.1 mol/L sodium hydroxide solution (A.3.4) or the standard volumetric 0.1 mol/L perchloric acid solution (A.3.5) as required to the combined sample solution. Transfer the resultant combined sample solution into a one-mark volumetric flask of suitable nominal

capacity (see table A.1). Make up to the mark with water and mix well.

A.5.8 To take further waste gas samples, place into each of the two absorbers the appropriate volume of absorption solution, replace the absorbers and proceed as specified above. In such cases ensure that greater volumes of ambient air do not enter the particle filter or the sampling probe when the absorbers are disconnected from the sampling train.

NOTE 1 Experience shows that a temperature in the particle filter within the range 150 °C to 200 °C is sufficient.

NOTE 2 If a wet-gas meter is used in the sampling train, prior to sampling a volume of the waste gas under investigation should be passed through the sampling train that contains a mass of carbon dioxide (CO₂) greater than that needed to saturate the sealing liquid in the wet-gas meter with carbon dioxide. Experience shows that a volume of waste gas under investigation within the range 0.05 m³ and 0.1 m³ is sufficient. Furthermore, the temperature of the sealing liquid in the wet-gas meter should not significantly differ from the gas passing through.

Table A.1 Variation of the volume of waste gas to pass through the sampling train, joint sample solution of portion of the treated joint sample solution to be titrated and the resultant volume of Ba (ClO₄)₂ solution related to the range of mass concentration of sulfur dioxide

Expected mass concentration of sulfur dioxide	Capacity of the absorbers	Volume of absorption solution to be placed into each of two absorbers	Volume of the waste gas to pass through the sampling train	Volume of the treated joint sample solution	Portion of the treated joint sample solution to be titrated	Volume of Ba (ClO ₄) ₂ solution used for the titration of the portion
mg/m ³	ml	ml	m ³	ml	ml	ml
30 to 100	100	40	0.100	100	20	1.875 to 6.25
100 to 500	100	40	0.060	100	20	3.75 to 18.75
500 to 1 000	250	80	0.060	250	20	7.5 to 15.0
1 000 to 2 000	250	80	0.030	250	20	7.5 to 15.0
2 000 to 5 000	250	80	0.030	250	20	15.0 to 37.5
2 000 to 5 100	250	80	0.030	250	10 ^{a)}	7.5 to 18.75
Note ^{a)} The portion of the joint sample solution to be titrated is diluted to 20 ml with distilled water.						

A.6 Procedure

A.6.1 Test of absorption efficiency

Place the appropriate volume of absorption solution (see table A.1) into each of two identical absorbers. Assemble the apparatus specified in A.4.1 to obtain a sampling train. Use spherical ground glass joints upstream of the second absorber.

By reference to table A.1 use a sampling period that results in the absorption of about 0.5 mg of sulfur dioxide per 1 ml of the absorption solution in the first absorber. Carry out sampling as specified in A.5.

Calculate the absorption efficiency by dividing the volume of the standard volumetric 0.005 mol/L barium perchlorate solution (A.3.3) used for titration of the sample solution in the first absorber by the sum of the volumes of the standard volumetric 0.005 mol/L barium perchlorate solution used for titration of the sample solutions in the first and second absorbers.

The absorption efficiency shall be at least 0.95. Absorbers that do not fulfil these requirements shall not be used.

A.6.2 Determination

Transfer the portion of the treated combined sample solution specified in table A.1 into a conical flask of suitable nominal capacity and add 80 ml of 2-propanol (A.3.1) plus four drops of the Thorin solution (A.3.6) and mix well. Record the ratio, f_v , of the volume of the treated combined sample solution to the volume of the portion taken.

Titrate the solution obtained with the standard volumetric 0.005 mol/L barium perchlorate solution (A.3.3) to the change in colour from orange-yellow to permanent light pink, using either a burette with a spindle-valve or an automatic titrator, and record the volume needed for titration of this solution. Repeat this procedure and, if either endpoint is indistinct, a third time and calculate the mean, V_1 of these values.

NOTE : In some types of light, in which the change in colour from orange-yellow to permanent light pink is very difficult to observe, as for example in fluorescent lighting and direct sunlight, titration should be carried out under a daylight lamp or, preferably, using a photometer with a glass fibre-optic light guide capable of measuring the transmission of liquids at a wavelength of 520 nm, and an automatic titrator.

A.6.3 Preparation of blank solution

Prepare a blank solution by adding 80 ml of 2-propanol (A.3.1) and four drops of the Thorin solution (A.3.6) to a portion of the absorption solution, the volume of which corresponds to the volume of the selected portion of the treated joint sample solution, and proceed as specified above. The blank value, V_2 , shall usually be evaluated each time the absorption solution is prepared.

NOTE : If the mass concentration of sulfur dioxide present in the waste gas under investigation is much greater than the detection limit of the method (see A.7.2.1), then the evaluation of the blank values may be dispensed with.

A.6.4 Interferences

A.6.4.1 Sulfur trioxide

Sulfur trioxide (SO_3) is absorbed in the absorption solution resulting in the formation of sulfuric acid. In many of the waste gases under investigation, however, sulfur trioxide is present in small quantities and, in such circumstances, regard the effect on the

titration reading of sulfur trioxide as negligible. For example, in waste gases from combustion facilities the mass concentration of sulfur trioxide is smaller than 0.05 times the mass concentration of sulfur dioxide. In cases in which the mass concentration of sulfur dioxide is to be determined separately from the mass concentration of sulfur trioxide present in the waste gases under investigation, use a method other than that specified in this Standard.

A.6.4.2 Volatile sulfates

Volatile sulfates that, under the conditions of sampling, form sulfate ions in the absorption solution can interfere.

A.6.4.3 Anions

Anions that are absorbed by the absorption solution and that also form sparingly soluble salts with barium ions at pH 3.5 can interfere.

NOTE : Interferences by inorganic gaseous chlorine and fluorine compounds and stable oxides are not expected in normal flue gases. In exceptional cases (extremely high flue-gas temperature) interferences by volatile sulfates and volatile salts of polyvalent metal cations can take place.

A.6.4.4 Volatile salts of polyvalent metal cations

Those which respond to the indicator Thorin. The metal cations, however, can be removed from the treated joint sample solution by percolating the latter through a cation exchanger (see NOTE of A.6.4.3).

A.7 Expression of results

A.7.1 Calculation

Calculate the mass concentration of sulfur dioxide ρ (SO₂) (unit : mg/m³) present in the waste gas sample at the state defined by $p_r = 101.32$ kPa and $T_r = 273.15$ K, related to the dry waste gas, by using the equation.

$$\rho(\text{SO}_2) = f_A \times f_V \times \frac{V_1 - V_2}{V_3} \times \frac{101.32 \times (273.15 + \theta_i)}{273.15 \times (p - p_{\text{H}_2\text{O}})}$$

where, f_A : the sulfur dioxide equivalent ($\mu\text{g/ml}$) of 0.005 mol/L barium perchlorate solution used

f_V : the ratio of the volume of the treated joint sample solution to the volume of the portion titrated

p : the atmospheric pressure (kPa) at the sampling stage

$p_{\text{H}_2\text{O}}$: the partial pressure of water vapor (kPa) at θ_i °C ; if a dry gas meter is used, the partial pressure of water is 0 kPa

V_1 : the mean (ml) of the values of the volume of the standard volumetric 0.005 mol/L barium perchlorate solution (A.3.3) used for titrating the portion (see table A.1) of the treated joint sample solution

V_2 : the mean (ml) of the values of the volume of the standard

volumetric 0.005 mol/L barium perchlorate solution (A.3.3) used for titrating the blank solutions

V_3 : the volume (m³) of the waste gas sample

θ_i : the temperature (°C) of the volume of the waste gas under investigation passing through the gas meter

A.7.2 Performance characteristics

A.7.2.1 Lower detection limit

The lower detection limit of the method is 0.72 mg/m³, when passing the volume of 0.09 m³ of the waste gas under investigation through the sampling train and selecting the portions specified in table A.1.

A.7.2.2 Standard deviation under repeatable conditions

Based on the mass concentrations of sulfur dioxide found to be present in identical volumes, for one and the same interval of time, of a waste gas from much the same measurement point, the standard deviations under repeatable conditions given in table A.2 have been evaluated :

Table A.2 Standard deviation under repeatable conditions

Mean of mass concentration of sulfur dioxide (mg/m ³)	Number n of pairs of values N	Standard deviation of the series of measurements based on n pairs of values (mg/m ³)
16	10	4.4
221	35	7.1
470	10	17
2 000	28	74

A.8 Test report

The test report shall include at least the following information :

- reference to this Standard ;
- identification of the sample ;
- description of plant and process ;
- plant operating conditions ;
- location of the measurement plane ;
- location of the measurement point in the measurement plane ;
- changes in the plant operations during sampling, for example burner changes ;
- any operation not specified in this Standard or regarded as optional ;
- the test result ;
- date and time of sampling.

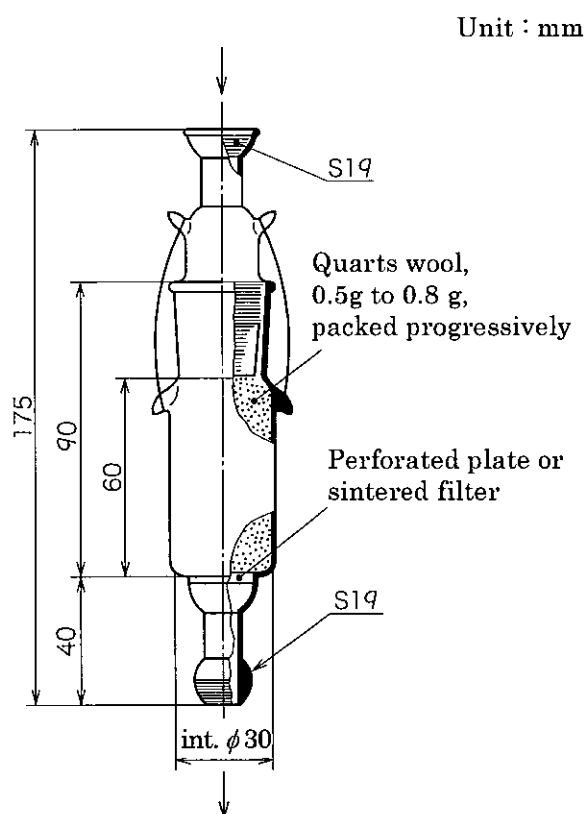


Figure A.1 Example of
a particle filter

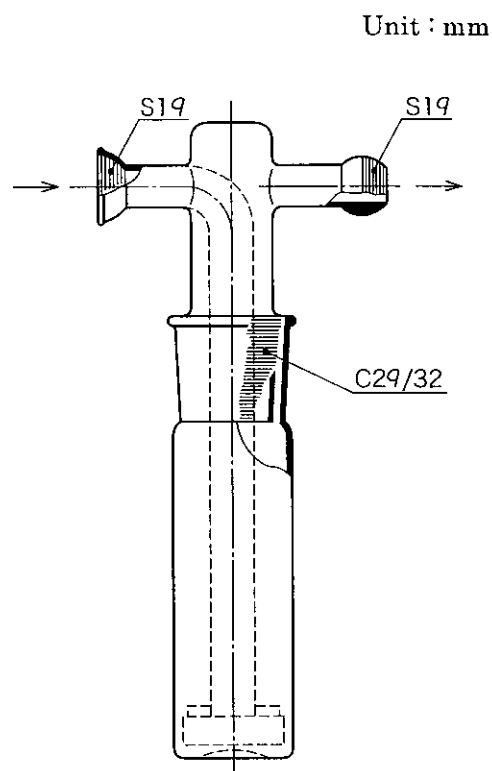


Figure A.2 Absorption bottle

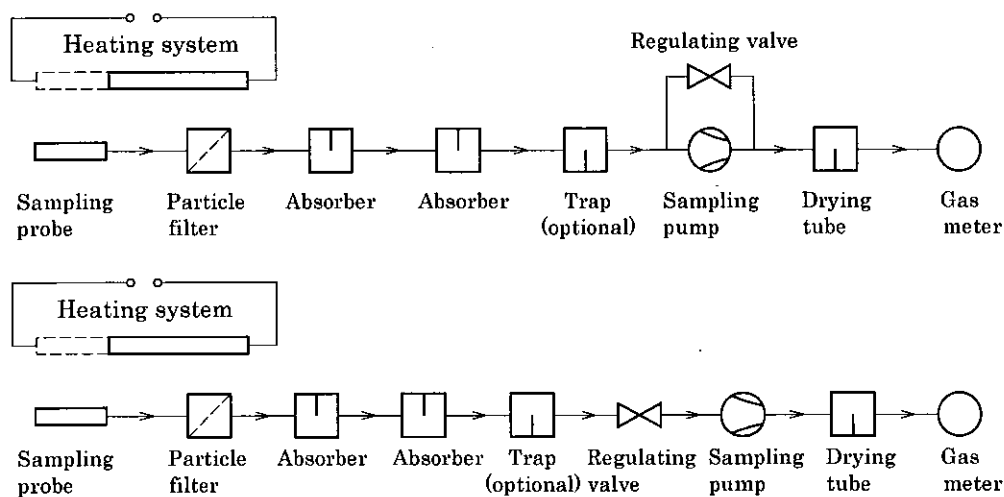


Figure A.3 Schematic representation of sampling trains for
the determination of the concentration of sulfur
dioxide in waste gases from stationary sources

Annex JA (normative)

Turbidimetry (light scattering method)

JA.1 General

This Annex specifies the analysis method of sulfur oxides by turbidimetry. This method is influenced when suspended matters are contained in the sample solution for analysis, therefore this is applicable when the influence thereof are ignorable or removable.

JA.2 Summary of analysis method

The summary of analysis method shall be in accordance with table JA.1.

Table JA.1 Summary of analysis method

Outline	Sampling	Determination range vol ppm (mg/m ³)	Applicable condition
After absorption of sulfur oxides in sample gas into hydrogen peroxide solution to make sulfuric acid, glycerol solution and sodium chloride solution are added and further barium chloride is added. The absorbance (420 nm) of barium sulfate generated in white turbidity is measured.	Absorbing bottle method Absorption solution : hydrogen peroxide solution (1 + 9) Liquid amount : 50 ml ^{a)} × 2 bottles Standard sampling volume : 20 L	5 to 300 ^{b)} (14 to 850) ^{b)}	According to JA.1.
Notes ^{a)} Liquid amount for absorption when using absorbing bottle (250 ml in capacity). ^{b)} In the case of employing the sample solution for analysis prepared by diluting 100 ml of absorption solution through which the sample gas has been passed to 250 ml. The determination range shown in this column is obtained according to the standard sampling volume of sample gas, the amount of sample solution for analysis and the optimum range of the working curve. For the concentration over the determination range, the sample solution for analysis shall be diluted within the determination range.			

JA.3 Sampling method of sample gas

JA.3.1 Sampling of sample gas

The sampling of sample gas shall be in accordance with clause 5.

JA.3.2 Reagent and preparation of absorption solution

- a) **Hydrogen peroxide** That specified in JIS K 8230.
- b) **Absorption solution [hydrogen peroxide solution (1 + 9)]** Take 100 ml of hydrogen peroxide and add 900 ml of water to it. Store it in dark and cold place.

JA.3.3 Implements and apparatus

In accordance with 5.3.

JA.3.4 Sampling procedure

In accordance with 5.4.

JA.3.5 Preparation of sample solution for analysis

After sulfur oxides in flue gas are collected in the absorption solution of JA.3.2 b), it shall be prepared according to the same procedure of 6.2.

JA.3.6 Calculation of sampling amount of sample gas

In accordance with 5.5.

JA.4 Analysis method**JA.4.1 Reagents and preparation of reagent solutions****JA.4.1.1 Reagents**

- a) **Hydrochloric acid** That specified in JIS K 8180.
- b) **Sulfuric acid** That specified in JIS K 8951.
- c) **Barium chloride** That specified in JIS K 8155 ground to 500 μm to 710 μm in size.
- d) **Sodium chloride** That specified in JIS K 8150.
- e) **Glycerol** That specified in JIS K 8295.

JA.4.1.2 Preparation of reagent solution

- a) **Sodium chloride solution** Dissolve 240 g of sodium chloride in 20 ml of hydrochloric acid, add water to make the total 1 L, and filtrate it with the filter paper of Class 5C.
- b) **Glycerol solution (1 + 1)** Mix 1 volume of glycerol and 1 volume of water.
- c) **Sulfate ion reference solution (SO_4^{2-} : 0.2 mg/ml)** Prepare in accordance with 7.1.2 b) 4), 1 ml of which corresponds to 0.046 6 ml of sulfur oxides ($\text{SO}_2 + \text{SO}_3$).

JA.4.2 Procedure for determination

The procedure for determination shall be as follows.

- a) Take 50 ml of sample solution for analysis prepared in 6.2 into each of two 100 ml beakers.
- b) Add 10 ml of glycerol solution (1 + 1) and 5 ml of sodium chloride solution into each of two beakers, and stir them sufficiently by a magnetic stirrer.
- c) Add 0.3 g of barium chloride into one beaker of them, stir it for 1 min, allow it to stand for 4 min, and again stir for 15 s.
- d) Take a part of the solution to which barium chloride has been added into an absorption cell, and measure the absorbance in the vicinity of 420 nm wavelength

while setting the solution without barium chloride to be the contrast solution.

For the sample containing sulfur oxides of 30 vol ppm or under in concentration, an absorption cell with 50 mm optical path length shall be used.

- e) Making use of the working curve prepared in JA.5, obtain the amount of sulfate ion (mg).
- f) Take 100 ml of absorption solution in 250 ml volumetric flask, add water up to the marked line, take 50 ml of the solution into each of two 100 ml beakers, and obtain the blank test value according to the procedures from b) to e).

JA.5 Preparation of working curve

The preparation of working curve shall be as follows.

- a) Take stepwise several amounts from 5.0 ml to 25.0 ml of sulfate ion reference solution (SO_4^{2-} : 0.2 mg/ml) into 100 ml beakers, add water respectively to make the total 50 ml. Similarly to JA.4.2 a), take sulfate ion reference solution into two beakers by the same amount, and carry out the procedures from b) to f) of JA.4.2 respectively.

When the absorption cell of 50 mm is used, weigh out 0.5 ml to 5.0 ml of sulfate ion reference solution.

- b) Carry out the procedures from b) to d) of JA.4.2, and prepare the relation curve plotting the relation between the amount (mg) of sulfate ion and absorbance.

The working curve covering the vicinity of concentration to be measured, should preferably be prepared in parallel with the analysis of sample every time when making analysis.

JA.6 Calculation

Calculate the concentration of sulfur oxides in the sample gas according to the following formulae.

$$C_v = \frac{0.233 \times (a - b) \times \frac{250}{50}}{V_s} \times 1\,000$$

$$C_w = \frac{0.667 \times (a - b) \times \frac{250}{50}}{V_s} \times 1\,000$$

$$C_w = C_v \times 2.86$$

- where,
- C_v : volume fraction of sulfur oxides in sample gas (vol ppm)
 - C_w : mass concentration of sulfur oxides in sample gas expressed as sulfur dioxide (mg/m^3)
 - a : amount of sulfate ion (SO_4^{2-}) obtained in JA.4.2 e) (mg)
 - b : amount of sulfate ion (SO_4^{2-}) obtained by blank test in JA.4.2 f) (mg)

- V_s : sample gas volume sampled under standard condition
calculated in 5.5 (L)
(V_{SD} for dry-base gas volume and V_{SW} for wet-base gas
volume)
- 0.233 : volume of sulfur oxides ($SO_2 + SO_3$) equivalent to 1 mg of
sulfate ion (SO_4^{2-}) (ml) (standard condition)
- 0.667 : mass of sulfur dioxide (SO_2) equivalent to 1 mg of sulfate
ion (SO_4^{2-}) (mg)
- 2.86 : mass concentration of sulfur dioxide (SO_2) equivalent to
1 vol ppm of sulfur oxides (mg/m^3) (according to 64.06/22.41)

Annex JB (normative)

Neutralization titrimetry

JB.1 General

This Annex specifies the analysis method of sulfur oxides by neutralization titrimetry.

This method is influenced when other acidic gas or ammonia coexists in the sample gas, therefore this is applicable when the influence thereof are ignorable or removable.

JB.2 Summary of analysis method

The summary of analysis method shall be in accordance with table JB.1.

Table JB.1 Summary of analysis method

Outline	Sampling	Determination range vol ppm (mg/m ³)	Applicable condition
After absorption of sulfur oxides in sample gas into hydrogen peroxide solution to make sulfuric acid, it is titrated with sodium hydroxide solution by using mixed solution of methyl red-methylene blue as an indicator.	Absorbing bottle method Absorption solution : hydrogen peroxide solution (1 + 9) Liquid amount : 50 ml ^{a)} × 2 bottles Standard sampling volume : 20 L	70 to 2 800 ^{b)} (200 to 8 000) ^{b)}	According to JB.1.
Notes ^{a)} Liquid amount for absorption when using absorbing bottle (250 ml in capacity). ^{b)} In the case of employing the sample solution for analysis prepared by diluting 100 ml of absorption solution through which the sample gas has been passed to 250 ml. The determination range shown in this column is obtained according to the standard sampling volume of sample gas, the amount of sample solution for analysis and the optimum range of the working curve. For the concentration over the determination range, the sample solution for analysis shall be diluted within the determination range.			

JB.3 Sampling method of sample gas

JB.3.1 Sampling of sample gas

The sampling of sample gas shall be in accordance with clause 5.

JB.3.2 Reagent and preparation of absorption solution

- a) **Hydrogen peroxide** That specified in JIS K 8230.
- b) **Absorption solution [hydrogen peroxide solution (1 + 9)]** Take 100 ml of hydrogen peroxide and add 900 ml of water to it. Store it in dark and cold place.

JB.3.3 Implements and apparatus

In accordance with 5.3.

JB.3.4 Sampling procedure

In accordance with 5.4.

JB.3.5 Preparation of sample solution for analysis

After sulfur oxides in flue gas are collected in the absorption solution of JB.3.2 b), it shall be prepared according to the same procedure of 6.2.

JB.3.6 Calculation of sampling amount of sample gas

In accordance with 5.5.

JB.4 Analysis method**JB.4.1 Reagents and preparation of reagent solutions****JB.4.1.1 Reagents**

- a) **Sodium hydroxide** That specified in JIS K 8576.
- b) **Sodium tetraborate decahydrate** That specified in JIS K 8866.
- c) **Soda lime** That specified in JIS K 8603.
- d) **Methyl red** That specified in JIS K 8896.
- e) **Methylene blue** That specified in JIS K 8897.
- f) **Ethanol (95)** That specified in JIS K 8102.
- g) **Amidosulfuric acid** Reference material for volumetric analysis specified in JIS K 8005.
- h) **Absorption solution [hydrogen peroxide solution (1 + 9)]** That prepared in JB.3.2 b).

JB.4.1.2 Preparation of reagent solutions

- a) **Methyl red-methylene blue mixed indicator (colour change point : pH 5.4)** Dissolve 0.10 g of methyl red in 100 ml of ethanol (95), dissolve 0.10 g of methylene blue in 100 ml of ethanol (95), and mix them equally in volume. Put it into a brown bottle, and store it in a dark and cold place.
- b) **0.05 mol/L sodium hydroxide solution** ^{1), 2)} Take 50 g of sodium hydroxide in a polyethylene bottle, add about 40 ml of water, shake it while cooling with running water, and allow it to stand for several days in a cold place after stoppered to prepare the saturated solution.

Take about 5 ml of supernatant (equivalent to 4 g of sodium hydroxide), and add carbon dioxide-free water to make the total 2 L. Put this solution in a polyethylene bottle, and store it with a soda lime tube.

Put water into a flask made of borosilicate, boil for 15 min, attach a stopper

equipped with a soda-lime tube, and use it after cooling.

Standardization shall be carried out as follows.

- Lightly crush amidosulfuric acid of JB.4.1.1 g) in an agate mortar, and dry it in a desiccator under reduced pressure of (2.0 kPa or less) for about 48 h. Weigh its 1.0 g to 1.2 g to the nearest 0.1 mg.
- Dissolve it in water, transfer it into a 250 ml volumetric flask, and add water up to the marked line.
- Take 25 ml of this solution into a 200 ml conical beaker, dilute it with water to about 100 ml, add 6 to 8 drops of methyl red-methylene blue mixed indicator, and titrate it with 0.05 mol/L sodium hydroxide solution previously prepared to find an end point where the colour of solution turns green from violet.

Instead of the titration using an indicator, a pH meter may be used for the titration. In this case, an end point shall be found around pH 5.4.

- Calculate the factor according to the following formula.

$$f' = \frac{m \times \frac{b}{100} \times \frac{25}{250}}{a' \times 0.004\ 854}$$

where, f' : factor of 0.05 mol/L sodium hydroxide solution
 m : amount of amidosulfuric acid taken (g)
 a' : amount of 0.05 mol/L sodium hydroxide solution consumed in titration (ml)
 b : purity of amidosulfuric acid (%)
 0.004 854 : amount of amidosulfuric acid equivalent to 1 ml of 0.05 mol/L sodium hydroxide solution (g)

Notes ¹⁾ Instead of 0.05 mol/L sodium hydroxide solution, 0.05 mol/L sodium tetraborate solution may be used. In this case, dissolve 9.5 g of sodium tetraborate decahydrate in water to make the total 1 L, and standardize it with amidosulfuric acid.

²⁾ If the sample gas gives high concentration of sulfur oxides, 0.1 mol/L sodium hydroxide solution is available. In this case, preparation and standardization shall be carried out according to those for 0.05 mol/L sodium hydroxide solution.

JB.4.2 Procedure for determination

The procedure for determination shall be as follows.

- a) Take v ml (generally 25 ml to 100 ml) out of the sample solution for analysis prepared in 6.2 into a 200 ml conical beaker.
- b) Add 6 to 8 drops of methyl red-methylene blue mixed indicator, titrate it with 0.05 mol/L sodium hydroxide solution using a 5 ml or 10 ml burette to find an end point where the colour of solution turns green from violet.
- c) Take 100 ml of absorption solution into a 250 ml volumetric flask, add water up to

the marked line, take the same amount as that of **a)** into a 200 ml conical beaker, and titrate according to **b)** to find the blank test value.

JB.5 Calculation

Calculate the concentration of sulfur oxides in the sample gas according to the following formulae.

$$C_V = \frac{0.560 \times (a - b) \times f \times \frac{250}{v}}{V_s} \times 1\,000$$

$$C_W = \frac{1.60 \times (a - b) \times f \times \frac{250}{v}}{V_s} \times 1\,000$$

$$C_W = C_V \times 2.86$$

- where,
- C_V : volume fraction of sulfur oxides in sample gas (vol ppm)
 - C_W : mass concentration of sulfur oxides in sample gas expressed as sulfur dioxide (mg/m³)
 - a : amount of 0.05 mol/L sodium hydroxide solution consumed in titration of **JB.4.2 b)** (ml)
 - b : amount of 0.05 mol/L sodium hydroxide solution consumed in blank test of **JB.4.2 c)** (ml)
 - f : factor of 0.05 mol/L sodium hydroxide solution
 - v : amount sample solution for analysis taken in **JB.4.2 a)** (ml)
 - V_s : sample gas volume sampled under standard condition calculated in **5.4** (L)
(V_{SD} for dry-base gas volume and V_{SW} for wet-base gas volume)
 - 0.560 : volume of sulfur oxides (SO₂ + SO₃) equivalent to 1 ml of 0.05 mol/L sodium hydroxide solution (ml) (standard condition)
 - 1.60 : mass of sulfur dioxide (SO₂) equivalent to 1 ml of 0.05 mol/L sodium hydroxide solution (mg)
 - 2.86 : mass concentration of sulfur dioxide (SO₂) equivalent to 1 vol ppm of sulfur oxides (mg/m³) (according to 64.06/22.41)

Annex JC (normative)

Simultaneous analysis method of sulfur oxides and hydrogen chloride by ion chromatography

JC.1 General

This Annex specifies the simultaneous analysis method of sulfur oxides and hydrogen chloride by ion chromatography.

JC.2 Summary of analysis method

The summary of analysis method shall be in accordance with table JC.1.

Table JC.1 Summary of analysis method

Outline	Sampling	Determination range vol ppm (mg/m ³)		Applicable condition
		Sulfur ox- ides	Hydrogen chloride	
After sulfur oxides and hydrogen chloride in sample gas are absorbed in hydrogen peroxide solution, it is injected in ion chromatograph and chromatogram is recorded.	Absorbing bottle method Absorption solution : hydrogen peroxide solution (1 + 99)	0.5 to 58 ^{c)} (1.4 to 165) ^{c)}	0.4 to 7.9 ^{c)} (0.7 to 12) ^{c)}	According to 7.1.1
	Liquid amount : 25 ml ^{a)} × 2 bottles or 50 ml ^{b)} × 2 bottles Standard sampling volume : 20 L	12 to 290 ^{d)} (35 to 829) ^{d)}	6.3 to 160 ^{d)} (10 to 260) ^{d)}	
Notes ^{a)} Liquid amount for absorption when using absorbing bottle (100 ml in capacity). ^{b)} Liquid amount for absorption when using absorbing bottle (250 ml in capacity). ^{c)} In the case of employing the sample solution for analysis prepared by diluting 50 ml of absorption solution through which the sample gas has been passed to 100 ml. The determination range shown in this column is obtained according to the standard sampling volume of the sample gas, the amount of sample solution for analysis and the optimum range of the working curve. When determining the concentration in this determination range or under, the concentrating column shall be used. ^{d)} In the case of employing the sample solution for analysis prepared by diluting 100 ml of absorption solution through which the sample gas has been passed to 250 ml. The determination range shown in this column is obtained according to the standard sampling volume of the sample gas, the amount of sample solution for analysis and the optimum range of the working curve. For the concentration over the determination range, the sample solution for analysis shall be diluted within the determination range.				

JC.3 Sampling method of sample gas

JC.3.1 Sampling of sample gas

The sampling of sample gas shall be in accordance with clause 5.

JC.3.2 Reagents and preparation of absorption solution

- a) **Hydrogen peroxide** That specified in JIS K 8230.
- b) **Absorption solution [hydrogen peroxide solution (1 + 99)]** In accordance with 5.2.2 b).

JC.3.3 Implements and apparatus

In accordance with 5.3.

JC.3.4 Sampling procedure

In accordance with 5.4.

JC.3.5 Preparation of sample solution for analysis

After sulfur oxides and hydrogen chloride in flue gas are collected in the absorption solution of JC.3.2 b), it shall be prepared according to the same procedure of 6.1.

JC.3.6 Calculation of sampling amount of sample gas

In accordance with 5.5.

JC.4 Analysis method**JC.4.1 Reagent and preparation of reagent solutions****JC.4.1.1 Reagent**

Sodium chloride Reference material for volumetric analysis specified in JIS K 8005.

JC.4.1.2 Preparation of reagent solutions

- a) **Eluant** In accordance with 7.1.2 b) 1).
- b) **Regeneration solution** In accordance with 7.1.2 b) 2).
- c) **Chloride ion reference solution (Cl^- : 1 000 mg/L)**
- d) **Chloride ion reference solution (Cl^- : 1 mg/L)** That whose traceability specified in the national measurement standard is secured. Or heat sodium chloride at about 600 °C for about 1 h beforehand and allow it to cool in a desiccator as it is. Weigh out 1.648 g thereof, dissolve it in a small amount of water, transfer it into a 1 000 ml of volumetric flask, and add water up to the marked line.
- e) **Chloride ion reference solution (Cl^- : 0.1 mg/ml)** Take 10 ml of chloride ion reference solution (Cl^- : 1 mg/L) into a 100 ml volumetric flask and add water up to the marked line. Prepare it at every service.
- f) **Mixed reference solution of sulfate ion and chloride ion [SO_4^{2-} : 0.05 mg, Cl^- : 0.01 mg)/ml]** Take 25 ml of sulfate ion reference solution (SO_4^{2-} : 0.2 mg/ml) of 7.1.2 b) 4), and 10 ml of chloride ion reference solution (Cl^- : 0.1 mg/ml) of e) into a 100 ml volumetric flask and add water up to the marked line. Prepare it at every service.

JC.4.2 Apparatus**JC.4.2.1 Ion chromatograph**

In accordance with 7.1.3.

JC.4.3 Procedure for determination

The procedure for determination shall be as follows.

- Carry out the procedures of a) and b) of 7.1.4.
- Obtain the peak areas or the peak heights for the peaks corresponding to sulfate ion and chloride ion on the chromatogram.
- Obtain the sulfate ion concentration (mg/ml) and the chloride ion concentration (mg/ml) in the sample solution for analysis according to the working curves.
- Take 50 ml of absorption solution prepared in 5.2.2 b) into a 100 ml volumetric flask, add water up to the marked line, take the same amount as introduced in 7.1.4 b), carry out the procedures in accordance with a) to c), and obtain the blank test values (mg/ml) of sulfate ion and chloride ion in the absorption solution. When the absorbing bottle (250 ml in capacity) is used, take 100 ml of absorption solution into a 250 ml volumetric flask, add water up to the marked line, and carry out the same procedures.

JC.5 Preparation of working curve

Use the mixed reference solution of sulfate ion–chloride ion [$(\text{SO}_4^{2-} : 0.05 \text{ mg}, \text{Cl}^- : 0.01 \text{ mg})/\text{ml}$] of JC.4.1.2 f), and prepare the working curves in accordance with 7.1.5.

JC.6 Calculation

The concentrations of sulfur oxides and hydrogen chloride in the sample gas shall be calculated according to the following formulae.

- Calculate the concentration of sulfur oxides in the sample gas according to formula (5) to formula (7) in 7.1.6.
- Calculate the concentration of hydrogen chloride in the sample gas according to the following formulae.

$$C_v = \frac{0.632 \times (a - b) \times v}{V_s} \times 1\,000$$

$$C_w = \frac{1.03 \times (a - b) \times v}{V_s} \times 1\,000$$

$$C_w = C_v \times 1.63$$

where, C_v : volume fraction of hydrogen chloride in sample gas (vol ppm)

C_w : mass concentration expressed as hydrogen chloride in sample gas (mg/m^3)

a : chloride ion concentration obtained in JC.4.3 c) (mg/ml)

b : chloride ion concentration obtained by blank test in JC.4.3 d) (mg/ml)

v : volumetric flask capacity (100 for 100 ml and 250 for 250 ml) (ml)

- V_s : sample gas volume sampled under standard condition
calculated in 5.4 (L)
(V_{SD} for dry-base gas volume and V_{SW} for wet-base gas
volume)
- 0.632: volume of hydrogen chloride (HCl) equivalent to 1 mg of
chloride ions (Cl^-) (ml)
(standard condition)
- 1.03: mass of hydrogen chloride (HCl) equivalent to 1 mg of
chloride ions (Cl^-) (mg)
- 1.63: mass concentration of hydrogen chloride equivalent to 1
vol ppm of hydrogen chloride (mg/m^3) (according to 36.46/
22.41)

Annex JD (normative)

Simultaneous analysis method of sulfur oxides, hydrogen chloride and nitrogen oxides by ion chromatography

JD.1 General

This Annex specifies the simultaneous analysis method of sulfur oxides, hydrogen chloride and nitrogen oxides in flue gas by ion chromatography. Hydrogen chloride and nitrogen oxides can be measured simultaneously with sulfur oxides by ion chromatography in 7.1. In this case, the sample gas is sampled and the sample solution for analysis is prepared in accordance with 6.1 or 6.2 of JIS K 0104, then these are determined according to the following method. This method is influenced by reductive gas such as sulfide in high concentration, therefore this is applicable when the influence thereof are ignorable or removable.

JD.2 Summary of analysis method

The summary of analysis method shall be in accordance with table JD.1.

Table JD.1 Summary of analysis method

Outline	Sampling	Determination range vol ppm (mg/m ³)		
		Sulfur oxides	Hydrogen chloride	Nitrogen oxides
After absorption of sulfur oxides, hydrogen chloride and nitrogen oxides in sample gas into hydrogen peroxide solution in vacuum flask or syringe, it is introduced into ion chromatograph and chromatogram thereof is recorded.	Vacuum flask method Absorption solution : hydrogen peroxide solution (1 + 99) Liquid amount : 20 ml Standard sampling volume : 1 L	5 to 1 100 ^{c)} (14 to 3 100) ^{c)}	4 to 630 ^{c)} (6 to 1 000) ^{c)}	4 to 1 400 ^{c)} (8 to 2 800) ^{c)}
	Syringe method Absorption solution : hydrogen peroxide solution (1 + 99) Liquid amount : 20 ml ^{a)} Standard sampling volume : 400 ml ^{a)} 150 ml ^{b)}	24 to 5 800 ^{c)} (69 to 16 000) ^{c)}	16 to 3 100 ^{c)} (26 to 5 000) ^{c)}	19 to 7 200 ^{c)} (39 to 14 000) ^{c)}
NOTE : This method is the same as that of Annex JA of JIS K 0104.				
Notes ^{a)} Sample volume of sample gas when using syringe (500 ml in capacity).				
^{b)} Sample volume of sample gas when using syringe (200 ml in capacity).				
^{c)} In the case of employing the sample solution for analysis prepared by diluting 20 ml of absorption solution through which the sample gas has been passed to 100 ml. The determination limit shown in this table can be made lower by using the concentrating column.				

JD.3 Sampling method of sample gas**JD.3.1 Sampling of sample gas**

The sampling of sample gas shall be in accordance with clause 6 of JIS K 0104.

JD.3.2 Reagents and preparation of reagent solutions

Reagents specified in 7.1.2 and used in the ion chromatography, and also specified in 6.1.1 of JIS K 0104 shall be used.

- a) **Absorption solution [hydrogen peroxide solution (1 + 99)]** Take 10 ml of hydrogen peroxide and add 990 ml of water. Store it in dark and cold place.
- b) **Implements and apparatus** In accordance with 6.1.2 or 6.2.2 of JIS K 0104.
- c) **Sampling procedure** In accordance with 6.1.3 or 6.2.3 of JIS K 0104.
- d) **Preparation of sample solution for analysis** After collecting sulfur oxides, hydrogen chloride and nitrogen oxides in flue gas simultaneously into the absorption solution of a), prepare it in accordance with 6.1.4 c) or 6.2.4 c) of JIS K 0104.
- e) **Calculation of sampling volume of sample gas** In accordance with 6.1.5 or 6.2.5 of JIS K 0104.

JD.4 Analysis method**JD.4.1 Reagents and preparation of reagent solutions****JD.4.1.1 Reagents**

The reagents specified in 7.1.2 a), 7.1.2.1 of JIS K 0107 and 7.3 of JIS K 0104 shall be used, unless otherwise specified.

- a) **Chloride ion reference solution (Cl^- : 1 000 mg/L)**
- b) **Nitrate ion reference solution (NO_3^- : 1 000 mg/L)**
- c) **Nitrite ion reference solution (NO_2^- : 1 000 mg/L)**

JD.4.1.2 Preparation of reagent solutions

The preparation of reagent solutions specified in 7.1.2 b), 7.1.2.2 of JIS K 0107 and 7.3.1 of JIS K 0104 shall be used, unless otherwise specified.

- a) **Eluant** In accordance with 7.1.2 b) 1).
- b) **Regeneration solution (removal solution)** In accordance with 7.1.2 b) 2).
- c) **Chloride ion reference solution (Cl^- : 1 mg/ml)** In accordance with JC.4.1.2 d).
- d) **Chloride ion reference solution (Cl^- : 0.1 mg/ml)** In accordance with JC.4.1.2 e).
- e) **Nitrate ion reference solution (NO_3^- : 1 mg/ml)** In accordance with 7.3.1 b) 3) of JIS K 0104.
- f) **Nitrate ion reference solution (NO_3^- : 0.1 mg/ml)** In accordance with 7.3.1 b) 4) of JIS K 0104.

- g) Nitrite ion reference solution (NO_2^- : 1 mg/ml) In accordance with 7.3.1 b) 5) of JIS K 0104.
- h) Nitrite ion reference solution (NO_2^- : 0.1 mg/ml) In accordance with 7.3.1 b) 6) of JIS K 0104.
- i) Mixed reference solution of sulfate ion, chloride ion and nitrate ion [SO_4^{2-} : 0.05 mg, Cl^- : 0.01 mg, NO_3^- : 0.04 mg)/ml] Take 25 ml of sulfate ion reference solution (SO_4^{2-} : 0.2 mg/ml) of 7.1.2 b) 4), 10 ml of chloride ion reference solution (Cl^- : 0.1 mg/ml) of d), and 40 ml of nitrate ion reference solution (NO_3^- : 0.1 mg/ml) of f) accurately into a 100 ml volumetric flask, and add water up to the marked line. Prepare it at every service and use this solution by adequately diluting. When nitrite ion is coexisted in the sample solution for analysis, prepare the mixed reference solution to which a definite amount of nitrite ion reference solution (NO_2^- : 0.1 mg/ml) of h) is added.

JD.4.2 Apparatus

JD.4.2.1 Ion chromatograph

In accordance with 7.1.3.

JD.4.3 Procedure for determination

The procedure for determination shall be as follows.

- a) Carry out the procedures in accordance with 7.1.4.
- b) Introduce a definite amount (10 μl to 250 μl) of sample solution for analysis prepared in 6.1 with a sample introducing device into the ion chromatograph, and record its chromatogram.
- c) Obtain the peak areas or the peak heights for the peaks corresponding to sulfate ion, chloride ion and nitrate ion on the chromatogram.
- d) Obtain the concentration of sulfate ion (mg/ml), chloride ion (mg/ml) and nitrate ion (mg/ml) with the working curves prepared in JD.5 ¹⁾.
- e) Take 20 ml of absorption solution prepared in JD.3.2 a) into a 100 ml volumetric flask, add water up to the marked line, take the same amount as introduced in b), carry out the procedures in accordance with b) to d), and obtain the blank test value of sulfate ion (mg/ml), chloride ion (mg/ml) and nitrate ion (mg/ml). When nitrite ion is coexisted, obtain the concentration of nitrite ion (mg/ml).

Note ¹⁾ When nitrite ion is coexisted, obtain the concentration of nitrite ion (mg/ml).

JD.5 Preparation of working curve

Use the mixed reference solution of sulfate ion, chloride ion and nitrate ion [SO_4^{2-} : 0.05 mg, Cl^- : 0.01 mg, NO_3^- : 0.04 mg)/ml] of JD.4.1.2 i), and prepare the working curves in accordance with 7.1.5.

JD.6 Calculation

Calculate the concentrations of sulfur oxides, hydrogen chloride and nitrogen oxides in the sample gas according to the following formulae.

- Calculate the sulfur oxides concentration in the sample gas according to the formulae in 7.1.6.
- Calculate the hydrogen chloride concentration in the sample gas according to the formulae in JC.6 b) or JA.4.6 b) 2) of JIS K 0104.
- Calculate the nitrogen oxides concentration in the sample gas.

$$C_v = \frac{[0.361 \times (a_1 - b_1) + 0.487 \times (a_2 - b_2)] \times 100}{V_s} \times 10^6$$

$$C_w = \frac{[0.742 \times (a_1 - b_1) + 1.000 \times (a_2 - b_2)] \times 100}{V_s} \times 10^6$$

$$C_w = C_v \times 2.05$$

- where,
- C_v : volume fraction of nitrogen oxides in sample gas (vol ppm)
 - C_w : mass concentration of nitrogen oxides in sample gas expressed as nitrogen dioxide (mg/m³)
 - a_1 : nitrate ion concentration obtained in JD.4.3 d) (mg/ml)
 - b_1 : nitrate ion concentration obtained by blank test in JD.4.3 e) (mg/ml)
 - a_2 : nitrite ion concentration obtained in JD.4.3 d) (mg/ml) ²⁾
 - b_2 : nitrite ion concentration obtained by blank test in JD.4.3 e) (mg/ml) ²⁾
 - V_s : sample gas volume sampled under standard condition in calculated in 6.1.5 or 6.2.5 of JIS K 0104 (ml)
(V_{SD} for dry-base gas volume and V_{SW} for wet-base gas volume)
 - 100 : when total amount of sample solution for analysis is 100 ml (ml)
 - 0.361 : volume of nitrogen oxides equivalent to 1 mg of nitrate ion (NO₃⁻) (ml) (standard condition)
 - 0.487 : volume of nitrogen oxides equivalent to 1 mg of nitrite ion (NO₂⁻) (ml) (standard condition)
 - 0.742 : mass of nitrogen dioxides (NO₂) equivalent to 1 mg of nitrate ion (NO₃⁻) (mg)
 - 1.000 : mass of nitrogen oxides (NO₂) equivalent to 1 mg of nitrite ion (NO₂⁻) (mg)
 - 2.05 : mass concentration of nitrogen dioxides (NO₂) equivalent to 1 vol ppm of nitrogen oxides (mg/m³) (according to 46.01/22.41)

Note ²⁾ Where nitrite ion is not existed, it is ignorable.

Annex JE (informative)
Comparison table between JIS and corresponding International Standards

JIS K 0103 : 2011 <i>Methods for determination of sulfur oxides in flue gas</i>							ISO 7934 : 1989 <i>Stationary source emissions — Determination of the mass concentration of sulfur dioxide — Hydrogen peroxide/barium perchlorate/Thorin method</i>		ISO 11632 : 1998 <i>Stationary source emissions — Determination of mass concentration of sulfur dioxide — Ion chromatography method</i>	
(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause	(V) Justification for the technical deviation and future measures				
No. and title of clause	Content		No. of clause	Content			Classification by clause	Detail of technical deviation		
1 Scope	Methods for determination of sulfur oxides in flue gas are specified.	ISO 11632	1	Method for determination of mass concentration of sulfur dioxide emitted from combustion facilities and technical processes is specified.	Alteration	In JIS, sulfur oxides (SO ₂ + SO ₃) are the object. In ISO Standards, sulfur dioxide (SO ₂) is the object.	It is better to unify the expression method of objective component for measurement, however, sulfur oxides are used in JIS because of the relation to the Air Pollution Control Law. Standardization is further encouraged.			
3 General matters	General matters on analysis methods and reagents are specified.	—	—	—	Addition	General matters are added.	Standardization is further encouraged.			
4 Classification and summary of analysis methods	a) Chemical analysis method Table 1 Summary of ion chromatography	ISO 11632	3	Identical with JIS.	Identical					

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
4 Classification and summary of analysis methods (continued)	Table 1 Objective components in table 1 SO ₂ + SO ₃		3	Objective components SO ₂	Alteration	In JIS, sulfur oxides (SO ₂ + SO ₃) are the object; however in ISO, sulfur dioxide (SO ₂) is the object.	In JIS and ISO Standard, the same absorption solution is used and the same components are measured, therefore the substantial concentration is same. ISO Standard uses absorbed SO ₃ as interfering substance; however a part of SO ₃ is also absorbed, therefore the expression in JIS will be proposed to ISO.
	Table 1 Absorption solution, gas sampling volume, etc. at sampling are specified.	ISO 11632 ISO 9734	5, 6 5, 6	Almost identical with JIS.	Alteration	Shape and capacity of absorption bottle, concentration of absorption solution, amount of solution, sampling amount of gas are different.	Slight deviation and there is no hindrance to measurement.
	Table 1 Determination range of ion chromatography	ISO 11632	1	Almost identical with JIS.	Alteration	Expression of concentration is vol ppm and mg/m ³ in JIS; however mg/m ³ in ISO Standard.	Expression of gas concentration is different, the expression of concentration in JIS will be proposed to ISO because it is expressed by ppm in Air Pollution Control Law in Japan.

(I) Requirements in JIS		(II) International Standard number		(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content			No. of clause	Content	Classification by clause	Detail of technical deviation	
4 Classification and summary of analysis methods (concluded)	Table 1 Summary of arsenazo III method	—		—	—	Addition	In ISO Standard, Thorin method is specified.	Arsenazo III indicator is clearer in colour change at end point than Thorin indicator, therefore, proposal of Arsenazo III method will be submitted to ISO.
	b) Automated measuring method SO ₂ in flue gas is specified.	—		—	—	Addition		Automated measuring method is added in consideration of user's convenience. Automated measuring method of ISO 7935 is conforming to that in JIS B 7981.
5 Sampling method of sample gas	5.2 Reagents and preparation of absorption solution	ISO 11632		4	Almost identical with JIS.	Alteration	Sampling instrument and sampling method are different, but basically identical. Concentration of hydrogen peroxide solution is different; however, there is no difference in test results.	Basically same. Sampling method for sample gas is specified in JIS K 0095 which is conforming to ISO 10396. As concentration of sulfur oxides in flue gas is low in Japan, it can be sufficiently collected in H ₂ O ₂ (1 + 99). The addition of this absorption solution will be proposed to ISO, because the influence on separation column in the case of using H ₂ O ₂ (1 + 99) is smaller.

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
5 Sampling method of sample gas (concluded)	5.3 Implements and apparatus	ISO 11632	5	Almost identical with JIS.	Alteration	Construction of sampling apparatus of sample gas is almost same basically, and test results are rarely influenced.	Slight deviation and there is no hindrance to measurement.
	5.4 Sampling procedure		6	Almost identical with JIS.	Alteration	Gas sampling procedure is almost same in JIS and ISO Standard. Gas sampling volume and sampling period are different.	
	5.5 Sampling volume of sample gas		6	Almost identical with JIS.	Alteration	In JIS, volume is specified as 20 L and in ISO Standard as 30 L. Gas sampling volume is different; however, in JIS, it may be increased or decreased according to the concentration of sulfur oxide.	
6 Preparation of sample solution for analysis	6.1 Ion chromatography	—	6.8	Almost identical with JIS.	Alteration	Procedure is slightly different but basically same.	Basically same and there is no hindrance to measurement.
	6.2 Precipitation titrimetry (arsenazo III method)		—	—	Addition	Not specified in ISO Standard.	

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
7 Determination by chemical analysis 7.1 Ion chromatography	7.1.1 Applicable condition	ISO 11632	5.2.2	Almost identical with JIS.		In JIS, applicable condition is clearly described, and the interfering substance is clearly described in ISO.	Slight deviation and there is no hindrance to measurement.
	7.1.2 Reagents and preparation of reagent solutions		4.2	Almost identical with JIS.	Alteration Addition	Representative example of eluant is clearly described in JIS, and the direction of manufacturer is referred to in ISO. b) 1) No suppressor system in eluant in ISO Standard.	
	7.1.3 Implements and apparatus		5.2.2	Almost identical with JIS; however, suppressor system is mainly specified.	Selection	In ISO Standard, suppressor system is mainly specified; however, both of suppressor system and non-suppressor system may be used in JIS.	Both of suppressor system and non-suppressor system are diffused in Japan, and specified in detail, therefore JIS system will be proposed to ISO.
	7.1.4 Procedure for determination		7 7.1 7.2	Almost identical with JIS.	Deletion	In ISO Standard, operation to confirm absorption efficiency is specified.	Verification test for obtaining absorption efficiency is carried out.

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
7 Determination by chemical analysis 7.1 Ion chromatography (continued)	7.1.5 Preparation of working curve	ISO 11632	7.4	Almost identical with JIS.	Alteration	In JIS, relation curve between concentration of sulfate ion and peak area or peak height is prepared. In ISO Standard, at least three kinds of concentration and absorption solution (zero solution) are used and working curve is prepared.	Slight deviation and there is no hindrance to measurement.
	7.1.6 Calculation		8	Almost identical with JIS.	Alteration	In JIS, concentration of sulfur oxides is calculated. In ISO Standard, concentration of sulfur dioxide is calculated.	
						In Air Pollution Control Law in Japan, concentration of sulfur oxides is expressed in volume fraction (vol ppm); however, in ISO Standard, concentration of sulfur dioxide is expressed in mass concentration (mg/m ³). Therefore, conversion formula to calculate both units of concentration is shown.	

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
7 Determination by chemical analysis 7.1 Ion chromatography (concluded)	—	ISO 11632	7.3.1 7.3.2 7.3.3 7.3.4	Almost identical with JIS; however, interfering substances (sulfur trioxide, volatile sulfates, ammonia, contaminants) are specified.	Deletion	In JIS, sulfur trioxide and volatile sulfates are together collected and calculated as sulfur oxides.	Influences by interfering substances, etc. are not sufficiently clear, therefore, the adoption of them in JIS is put off.
7.2 Precipitation titrimetry (arsenazo III method)		—	—	—	Selection	In ISO Standard, Thorin method is specified (Annex A), in JIS, arsenazo III method is also available.	Indicator, titration agent are different; however, technical deviation is slight. Arsenazo III method will be proposed to ISO.
8 Determination of automated measuring method	Measurement by means of measuring instrument specified in JIS B 7981.	—	—	—	Addition	Conforming to ISO 7935. Automated measuring method is added.	Added in consideration of user's convenience. JIS B 7981 is conforming to ISO 7935.
9 Method to obtain analysis value of flue gas	Obtaining method of analysis value.	ISO 11632	9	Analysis is carried out twice about same sample solution for analysis every sampling, and mean value is rounded to two significant figures.	Addition	Not specified in ISO Standard.	Slight deviation and there is no technical deviation.

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
10 Record of analysis	Items to be recorded.	ISO 11632	10	Almost identical with JIS.	Alteration	Items are different: however, there is no large deviation.	Slight deviation and there is no technical deviation.
Annex A (normative) Precipitation titrimetry (Thorin method)	After absorption of sulfur dioxide in sample gas into hydrogen peroxide to make sulfuric acid, pH is adjusted to 3.5, titrate it with barium perchlorate solution using Thorin as indicator.	ISO 7934	Overall	Identical with JIS.	Identical	ISO 7934 is made Annex A.	
Annex JA (normative) Turbidimetry (light scattering method)	After absorption of sulfur oxides in sample gas into hydrogen peroxide solution to make sulfuric acid, glycerol solution and sodium chloride solution are added and further barium chloride is added, and absorbance of barium sulfate generated in white turbidity is measured.	—	—	—	Addition	Not specified in ISO Standard.	This is used as analytical method for low concentration of sulfur oxides in Japan and made Annex in consideration of user's convenience. Standardization is further encouraged.

(I) Requirements in JIS		(II) International Standard number		(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content			No. of clause	Content	Classification by clause	Detail of technical deviation	
Annex JB (normative) Neutralization titrimetry	After absorption of sulfur oxides in sample gas into hydrogen peroxide solution to make sulfuric acid, it is titrated with sodium hydroxide solution.	—		—	—	Addition	Not specified in ISO Standard.	This is used as analytical method for high concentration of sulfur oxides in Japan and made Annex in consideration of user's convenience.
Annex JC (normative) Simultaneous analysis method of sulfur oxides and hydrogen chloride by ion chromatography	After sulfur oxides and hydrogen chloride in sample gas are absorbed in hydrogen peroxide solution, it is injected in ion chromatograph and chromatogram is recorded.	—		—	—	Addition	Not specified in ISO Standard.	This is used as simultaneous analysis method of sulfur oxides and hydrogen chloride, and made Annex in consideration of user's convenience. Simultaneous analysis method of JIS will be propose to ISO.

(I) Requirements in JIS		(II) International Standard number		(III) Requirements in International Standards		(IV) Classification and details of technical deviation between JIS and the International Standards by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content			No. of clause	Content	Classification by clause	Detail of technical deviation	
Annex JD (normative) Simultaneous analysis method of sulfur oxides, hydrogen chloride and nitrogen oxides by ion chromatography	After absorption of sulfur oxides, hydrogen chloride and nitrogen oxides in sample gas into hydrogen peroxide solution in vacuum flask or syringe, it is introduced into ion chromatograph and chromatogram thereof is recorded.	—		—	—	Addition	Not specified in ISO Standard.	This is used as simultaneous analysis method of sulfur oxides, hydrogen chloride and nitrogen oxides, and made Annex in consideration of user's convenience. Standardization is further encouraged.

Overall degree of correspondence between JIS and International Standards (ISO 7934 : 1989, ISO 11632 : 1998): MOD

NOTE 1 Symbols in sub-columns of classification by clause in the above table indicate as follows:

- Identical : Identical in technical contents.
- Deletion : Deletes the specification item(s) or content(s) of International Standard.
- Addition : Adds the specification item(s) or content(s) which are not included in International Standard.
- Alteration : Alters the specification content(s) which are included in International Standard.
- Selection : Provides an alternative choice by adding the specification content(s) of equal status, which may be used as an alternative to that given in the original International Standard.

NOTE 2 Symbol in column of overall degree of correspondence between JIS and International Standards in the above table indicates as follows:

— MOD : Modifies International Standards.

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